



Decentralized co-design of distributed controllers and communication topologies for vehicular platoons: A dissipativity-based approach[☆]

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ARTICLE INFO

Article history:

Received 9 December 2023

Received in revised form 25 July 2024

Accepted 3 December 2024

Available online 30 January 2025

ABSTRACT

Vehicular platoons provide an appealing option for future transportation systems. Most of the existing work on platoons separated the design of the controller and its communication topology. However, it is advantageous to simultaneously design the platooning controller and the communication topology, especially in cases of platoon splitting and merging. Therefore, we propose a co-design framework for vehicular platoons that maintains both the compositionality of the controller and the string stability of the platoon. In order to solve the co-design problem for a platoon of vehicles, we first formulate a centralized linear matrix inequality (LMI) problem using the dissipativity properties of the vehicles. We then decompose it using Sylvester's criterion to obtain smaller decentralized LMI problems. These smaller problems can be solved sequentially at individual vehicles in the platoon. To ensure the feasibility of the formulated centralized/decentralized LMI problems, we introduce local controllers at each vehicle, along with a specially developed local controller design scheme. In the centralized/decentralized LMI problems, a locally derived LMI condition is also encoded to ensure the finite-gain L2 stability of the closed-loop platooning system. We show that this implies L2 weak string stability of the vehicular platoon. Finally, we develop a simulation framework² and conduct simulation studies to validate the proposed co-design method and its compositionality features.

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1. Introduction

Vehicular platoons are a promising solution for future transportation development. By arranging all vehicles in closely spaced groups, platoons lead to a decreased aerodynamic drag and better utilization of road infrastructures (Besselink & Johansson, 2017). In this way, it can significantly increase the highway capacity, enhance traffic safety, and reduce fuel consumption and exhaust emissions (Jia, Lu, Wang, Zhang, & Shen, 2015).

In vehicular platoons, there are many interesting control problems, such as designing controllers to achieve the desired inter-vehicle distances and speeds. Plenty of control methods have been proposed, such as linear control (e.g., PID (Fiengo, Lui,

Petrillo, Santini, & Tufo, 2019), LQR/LQG (Tavan, Tavan, & Hosseini, 2015; Wang, Su, & Wang, 2022), H_∞ control (Herman, Martinec, Hurák, & Šebek, 2014; Ploeg, Shukla, Van De Wouw, & Nijmeijer, 2013)) and nonlinear control approaches (e.g., model predictive control (MPC) (Chen, Wang, Alkim, & Van Arem, 2018), optimal control (Morbidì, Colaneri, & Stanger, 2013; Ploeg, Shukla, et al., 2013), consensus-based control (Syed, Yin, Pandya, Zhang, et al., 2012), sliding-mode control (SMC) (Guo, Wang, Liao, & Teo, 2016; Xiao & Gao, 2011), backstepping control (Chou, Tang, Lu, & Bayen, 2019; Zhu & Zhu, 2018) and intelligent control (Ji, He, Lv, Liu, & Wu, 2018; Li & Yu, 2010)).

In most of these existing works, it is usually assumed that the communication topology of the platoon is fixed. However, in many practical scenarios, the communication topology is dynamic, and vehicles can join or leave a platoon anytime. For example, dynamic changes, such as merging and splitting of the vehicles, frequently occur when ramp vehicles merge into main-lanes, long platoons pass intersections, and vehicles exit a platoon and make lane changes (Guo & Wen, 2015). This poses significant challenges in the controller design for platoons since we must simultaneously guarantee the designed controller's compositionality and the platoon's string stability after vehicles join/leave the corresponding formations. On the one hand, the compositionality of the controllers implies that the designed controllers for other

[☆] This work was supported in part by National Science Foundation under grants CNS-1830335 and IIS-2007949. The material in this paper was not presented at any conference. This paper was recommended for publication in revised form by Associate Editor Subhrakanti Dey under the direction of Editor Florian Dorfler.

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² Available in GitHub: <https://github.com/NDzsong2/Longitudinal-Vehicular-Platoon-Simulator.git>.

remaining vehicles in the platoon are not required to be re-designed, e.g., re-tuning the control parameters from scratch, when such maneuverings occur. On the other hand, the string stability of the platoons captures the uniform boundedness of the tracking errors as they propagate along the downstream direction of the platoons.

The study of the platooning control that enables merging and splitting is only recent. Methods, such as PID (Dasgupta, Raghuraman, Choudhury, Teja, & Dauwels, 2017), MPC (Goli & Eskandarian, 2019), and cooperative approaches (Morales & Nijmeijer, 2016), have been proposed in the literature. However, most of these methods are either heuristic or based on traditional formation control, which lacks formal string stability guarantees (Dasgupta et al., 2017; Goli & Eskandarian, 2019; Morales & Nijmeijer, 2016). Moreover, these methods require controllers to be re-designed or re-tuned for the remaining vehicles after a vehicle leaves or joins the platoon. In other words, these methods are not compositional and lack a formal string stability guarantee.

We are, therefore, motivated to study the platooning control that can simultaneously guarantee the designed controller's compositionality and the platoon's string stability when vehicles join/leave the corresponding formations. Furthermore, it is known that the communication topology of the platoons plays a significant role in the performance of the platoons, e.g., stability margin, communication cost, security, etc. However, the study of topology-related impacts on the platoon is very sparse. Existing work includes the design of innovative communication topologies (Orki & Arogeti, 2019; Yan, Du, He, & Li, 2022), the analysis of the influence of the communication topologies on the platoon properties (Hao & Barooah, 2012; Hao, Barooah, & Mehta, 2011; Pirani, Baldi, & Johansson, 2022; Zheng, Li, Wang, Cao, & Li, 2015) and the impacts on other aspects, e.g., control strategies (Wijnbergen & Besselink, 2020) and disturbances rejection (Liu, Besselink, Baldi, Yu, & Trentelman, 2021). The topology study is separated from the controller synthesis in these existing works. We believe that a better practice is to design the controller and the communication topology together, especially when vehicles join/leave the platoon, where the topology change will impact the designed controller for the platoon. Therefore, in this paper, we design the communication topology and the distributed controller simultaneously for the platoon, which is called a co-design problem.

Our solution to the co-design problem is based on our recent work on a dissipativity-based design framework for general networked systems (Welikala, Lin, & Antsaklis, 2023). Based on this framework, the platooning controller and topology co-design problem is formulated as a centralized Linear Matrix Inequality (LMI) problem, and a decentralized way to solve such an LMI problem is proposed using the techniques presented in our recent work (Welikala, Lin, & Antsaklis, 2022a). Unlike the existing methods, our proposed co-design method is resilient (compositional) concerning the joining/leaving of the vehicles in a platoon, i.e., there is no need to re-design the controllers for all the remaining vehicles but rather modify the controllers for those affected vehicles. It is also shown that the designed controller ensures the L_2 stability of the closed-loop platooning system, which further implies the L_2 weak string stability. Therefore, overall, the proposed co-design approach ensures compositionality as well as string stability.

It is worth noting that our prior work (Welikala et al., 2022a) and Welikala et al. (2023) have studied general linear and non-linear networked systems, respectively. In there, we have developed techniques for: (1) decentralization of analysis and control design, and (2) control and topology co-design, respectively. In contrast, in this paper, we consider the vehicular platooning application and present how the fundamental techniques developed in Welikala et al. (2022a, 2023) can be applied, while

addressing new application-specific challenges such as: (1) establishing string stability guarantees, (2) designing local controllers to support global controller and topology co-design, and (3) decentralization of LMIs involving non-network structured matrices.

This work is also loosely related to our previous work on distributed backstepping controllers for platoons (Song, Welikala, Antsaklis, & Lin, 2022). However, the novelty of this work lies in the following aspects:

- (1) By assuming all the followers have access to the leader's information, we propose a distributed platooning controller that can be designed centrally and decentrally (i.e., in an incremental/compositional manner) using corresponding LMIs.
- (2) For the closed-loop platoon system under this distributed controller, we propose an optimization-based topology synthesis approach to determine the control parameters for the controller, which also represent the weighted communication links between vehicles.
- (3) To support the feasibility of this controller-topology synthesis (both in the centralized and decentralized settings), we introduce local controllers at the vehicles along with a specifically developed local controller design scheme.
- (4) The resulting topology (controller with certain parameters) guarantees the internal and the L_2 weak string stability of the closed-loop platooning system and is also security-aware and robust to disturbances.

Regarding the above first point, it is worth noting that (1) the assumption that the followers have access to the leader's information is commonly used in the literature (e.g., see Dasgupta et al. (2017), di Bernardo, Falcone, Salvi, and Santini (2015), Huang, Li, Zhang, Liu, and Mi (2022)) as it is justified in instances like where the leader is virtual, the leader's state can be estimated locally using distributed observers, or the leader's state is broadcasted using a local communication network, (2) the proposed centralized co-design process, as opposed to the decentralized version, is not compositional and thus it cannot be used to handle merging and splitting scenarios of vehicular platoons efficiently, and (3) the proposed decentralized co-design process is a specifically developed extension of the centralized version.

This paper is organized as follows. Section 2 summarizes several important notations and preliminary concepts about dissipativity, networked systems, decentralization, and string stability (corresponding proofs for this section are given in the Appendix). The platooning problem is formulated in Section 3, followed by the centralized and decentralized co-design solutions for the platoon detailed in Sections 4 and 5, respectively. In Section 6, the simulation results are presented. Finally, a concluding remark is made in Section 7.

2. Preliminaries

2.1. Notations

The sets of real and natural numbers are denoted by $\mathbb{R}$ and $\mathbb{N}$, respectively. We define $\mathbb{N}_N \triangleq \{1, 2, \dots, N\}$ where $N \in \mathbb{N}$. An $n \times m$ block matrix A can be represented as $A = [A_{ij}]_{i \in \mathbb{N}_n, j \in \mathbb{N}_m}$ where A_{ij} is the $(i, j)^{\text{th}}$ block of A (for indexing purposes, either subscripts or superscripts may be used depending on the availability, i.e., $A_{ij} \equiv A^{ij}$). $[A_{ij}]_{j \in \mathbb{N}_m}$ represents a block row matrix and $\text{diag}([A_{ii}]_{i \in \mathbb{N}_n})$ represents a block diagonal matrix. We define $\{A^i\} \triangleq \{A_{ii}\} \cup \{A_{ij} : j \in \mathbb{N}_{i-1}\} \cup \{A_{ji} : j \in \mathbb{N}_{i-1}\}$. If $\Psi \triangleq [\Psi^{kl}]_{k, l \in \mathbb{N}_m}$ where each $\Psi^{kl} \triangleq [\Psi_{ij}^{kl}]_{i, j \in \mathbb{N}_n}$, the block element-wise form of Ψ is denoted as $\text{BEW}(\Psi) \triangleq [[\Psi_{ij}^{kl}]_{k, l \in \mathbb{N}_m}]_{i, j \in \mathbb{N}_n}$. For example, if $\Psi \triangleq [A \ B]$ with

$A \triangleq [A_{ij}]_{i,j \in \mathbb{N}_n}$ and $B \triangleq [B_{ij}]_{i,j \in \mathbb{N}_n}$, then $\text{BEW}(\Psi) \triangleq [A_{ij} \ B_{ij}]_{i,j \in \mathbb{N}_n}$.

The transpose of a matrix A is denoted by $A^\top$ and $(A^\top)^{-1} = A^{-\top}$. The zero and identity matrices are denoted by $\mathbf{0}$ and $\mathbf{I}$, respectively (dimensions will be clear from the context). A symmetric positive definite (semi-definite) matrix $A \in \mathbb{R}^{n \times n}$ is represented as $A = A^\top > \mathbf{0}$ ($A = A^\top \geq \mathbf{0}$). For notational convenience, we assume $A > \mathbf{0} \iff A = A^\top > \mathbf{0}$. The symbol $\star$ represents obvious conjugate matrices found inside symmetric block matrices. The symmetric part of a matrix A is $\mathcal{H}(A) \triangleq A + A^\top$. $\mathbf{1}_{\{\cdot\}}$ is the indicator function and $\mathbf{e}_{ij} \triangleq \mathbf{1}_{\{i=j\}}$. We use $\mathcal{K}$, $\mathcal{K}_\infty$ and $\mathcal{KL}$ to denote different classes of comparison functions (e.g., see Sontag and Wang (1995)). For a vector $x \in \mathbb{R}^n$, the Euclidean norm of it is given by $\|x\|_2 \triangleq |x| \triangleq \sqrt{x^\top x}$. All vectors are column vectors. For a time-dependent vector $x(t) \in \mathbb{R}^n$, the $\mathcal{L}_2$ and $\mathcal{L}_\infty$ norms of it are given by $\|x(\cdot)\| \triangleq \sqrt{\int_0^\infty |x(\tau)|^2 d\tau}$ and $\|x(\cdot)\|_\infty \triangleq \sup_{t \geq 0} |x(t)|$, respectively.

2.2. Dissipativity

Consider a control system

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t)), \\ y(t) &= h(x(t), u(t)), \end{aligned} \quad (1)$$

where $x(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}^q$, $y(t) \in \mathbb{R}^m$, and $f: \mathbb{R}^n \times \mathbb{R}^q \rightarrow \mathbb{R}^n$ and $h: \mathbb{R}^n \times \mathbb{R}^q \rightarrow \mathbb{R}^m$ are continuously differentiable. The equilibrium points of (1) are such that there exists a set $\mathcal{X} \subset \mathbb{R}^n$ where for every $x^* \in \mathcal{X}$, there is a unique $u^* \in \mathbb{R}^q$ that satisfies $f(x^*, u^*) = \mathbf{0}$ while both u^* and $y^* \triangleq h(x^*, u^*)$ being implicit functions of x^* . For the dissipativity analysis of (1) without the explicit knowledge of its equilibrium points, the *equilibrium-independent dissipativity* (EID) property (Arcak, 2022) defined next can be used. Note that it also includes the conventional dissipativity property (Willems, 1972), particularly when $\mathcal{X} = \{\mathbf{0}\}$ and for $x^* = \mathbf{0} (\in \mathcal{X})$, the corresponding $u^* = \mathbf{0}$ and $y^* = \mathbf{0}$.

Definition 1. The system (1) is EID under supply rate $s: \mathbb{R}^q \times \mathbb{R}^m \rightarrow \mathbb{R}$ if there exists a continuously differentiable storage function $V: \mathbb{R}^n \times \mathcal{X} \rightarrow \mathbb{R}$ satisfying: $V(x, x^*) > 0$ with $x \neq x^*$, $V(x^*, x^*) = 0$, and $\dot{V}(x, x^*) = \nabla_x V(x, x^*) f(x, u) \leq s(u - u^*, y - y^*)$, for all $(x, x^*, u) \in \mathbb{R}^n \times \mathcal{X} \times \mathbb{R}^q$.

This EID property can be specialized based on the used supply rate $s(\cdot, \cdot)$. In the sequel, we use the concept of X-EID property (Welikala et al., 2023), which is defined using a quadratic supply rate determined by the coefficients matrix $X = X^\top \in \mathbb{R}^{q+m}$.

Definition 2. The system (1) is X-EID (with $X \triangleq [X^{kl}]_{k,l \in \mathbb{N}_2}$) if it is EID under the quadratic supply rate:

$$s(u - u^*, y - y^*) \triangleq \begin{bmatrix} u - u^* \\ y - y^* \end{bmatrix}^\top \begin{bmatrix} X^{11} & X^{12} \\ X^{21} & X^{22} \end{bmatrix} \begin{bmatrix} u - u^* \\ y - y^* \end{bmatrix}.$$

Depending on the choice of the coefficients matrix X in the above-defined X-EID property, it can represent several properties of interest, as detailed in the following remark.

Remark 1. If the system (1) is X-EID with:

- (1) $X = \begin{bmatrix} \mathbf{0} & \frac{1}{2}\mathbf{I} \\ \frac{1}{2}\mathbf{I} & \mathbf{0} \end{bmatrix}$, then it is *passive*;
- (2) $X = \begin{bmatrix} -\nu\mathbf{I} & \frac{1}{2}\mathbf{I} \\ \frac{1}{2}\mathbf{I} & -\rho\mathbf{I} \end{bmatrix}$, then it is *strictly passive* (ν and ρ are the input feedforward and output feedback passivity indices (Welikala, Lin, & Antsaklis, 2022d), respectively);

$$(3) \ X = \begin{bmatrix} \gamma^2 \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{I} \end{bmatrix}, \text{ then it is } L_2\text{-stable } (\gamma \text{ is the } L_2\text{-gain, denoted as } L_2G(\gamma)).$$

In particular, if (1) is strictly passive with passivity indices ν and ρ as in the above Case 2, we also denote this as (1) being IF-OPF(ν, ρ) (Rahnama, Xia, & Antsaklis, 2016). This means the closed-loop system where (1) is connected with feedforward and feedback systems that are non-passive with input feedforward and output feedback passivity indices $-\nu$ and $-\rho$, respectively, will still be passive.

If the system (1) is linear time-invariant (LTI), a necessary and sufficient condition for it to be X-EID is given in the next proposition – as a linear matrix inequality (LMI) problem.

Proposition 1. The linear time-invariant (LTI) system

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t), \\ y(t) &= Cx(t) + Du(t), \end{aligned} \quad (2)$$

is X-EID if and only if there exists $P > \mathbf{0}$ such that

$$\begin{bmatrix} -\mathcal{H}(PA) + C^\top X^{22} C & -PB + C^\top X^{21} + C^\top X^{22} D \\ \star & X^{11} + \mathcal{H}(X^{12} D) + D^\top X^{22} D \end{bmatrix} \geq \mathbf{0}. \quad (3)$$

The following corollary considers a particular LTI system with a local controller (a configuration that we will encounter later on) and provides an LMI problem for synthesizing this local controller to enforce/optimize X-EID.

Corollary 1. The LTI system

$$\dot{x}(t) = (A + BL)x(t) + \eta(t), \quad (4)$$

is X-EID with $X^{22} < \mathbf{0}$ from external input $\eta(t)$ to state $x(t)$ if and only if there exists $P > \mathbf{0}$ and K such that

$$\begin{bmatrix} -(X^{22})^{-1} & P & \mathbf{0} \\ \star & -\mathcal{H}(AP + BK) & -I + PX^{21} \\ \star & \star & X^{11} \end{bmatrix} \geq \mathbf{0} \quad (5)$$

and $L = KP^{-1}$.

2.3. Networked systems

2.3.1. Configuration

Consider the networked system Σ shown in Fig. 1 comprised of N independent dynamic subsystems $\{\Sigma_i : i \in \mathbb{N}_N\}$ and a static interconnection matrix M that defines how the subsystems, an exogenous input signal $w(t) \in \mathbb{R}^r$ (e.g., disturbance) and an interested output signal $z(t) \in \mathbb{R}^l$ (e.g., performance) are interconnected with each other.

The dynamics of the subsystem Σ_i , $i \in \mathbb{N}_N$ are given by

$$\begin{aligned} \dot{x}_i(t) &= f_i(x_i(t), u_i(t)), \\ y_i(t) &= h_i(x_i(t), u_i(t)), \end{aligned} \quad (6)$$

where $x_i(t) \in \mathbb{R}^{n_i}$, $u_i(t) \in \mathbb{R}^{q_i}$, $y_i(t) \in \mathbb{R}^{m_i}$ and $u \triangleq [u_i^\top]_{i \in \mathbb{N}_N}^\top$, $y \triangleq [y_i^\top]_{i \in \mathbb{N}_N}^\top$. Analogous to (1), each subsystem Σ_i , $i \in \mathbb{N}_N$ is assumed to have a set $\mathcal{X}_i \subset \mathbb{R}^{n_i}$ where for every $x_i^* \in \mathcal{X}_i$, there is a unique $u_i^* \in \mathbb{R}^{q_i}$ that satisfies $f_i(x_i^*, u_i^*) = \mathbf{0}$ while both u_i^* and $y_i^* \triangleq h_i(x_i^*, u_i^*)$ being implicit functions of x_i^* . Moreover, each subsystem Σ_i , $i \in \mathbb{N}_N$ is also assumed to be X_i -EID where $X_i = X_i^\top \triangleq [X_i^{kl}]_{k,l \in \mathbb{N}_2}$ (see Definition 2).

On the other hand, the interconnection matrix M and the corresponding interconnection relationship are given by:

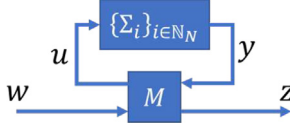
$$\begin{bmatrix} u \\ z \end{bmatrix} = \begin{bmatrix} M_{uy} & M_{uw} \\ M_{zy} & M_{zw} \end{bmatrix} \begin{bmatrix} y \\ w \end{bmatrix} \equiv M \begin{bmatrix} y \\ w \end{bmatrix}. \quad (7)$$

Note that we assume $w \triangleq [w_i^\top]_{i \in \mathbb{N}_N}^\top$ and $z \triangleq [z_i^\top]_{i \in \mathbb{N}_N}^\top$, where each $w_i \in \mathbb{R}^{r_i}$ and $z_i \in \mathbb{R}^{l_i}$ respectively represent an exogenous disturbance input and an interested performance output corresponding

$$\begin{bmatrix} \mathbf{X}_p^{11} & \mathbf{0} & L_{uy} & L_{uw} \\ \mathbf{0} & -\mathbf{Y}^{22} & -\mathbf{Y}^{22} M_{zy} & -\mathbf{Y}^{22} M_{zw} \\ L_{uy}^\top & -M_{zy}^\top \mathbf{Y}^{22} & -L_{uy}^\top \mathbf{X}^{12} - \mathbf{X}^{21} L_{uy} - \mathbf{X}_p^{22} & -\mathbf{X}^{21} L_{uw} + M_{zy}^\top \mathbf{Y}^{21} \\ L_{uw}^\top & -M_{zw}^\top \mathbf{Y}^{22} & -L_{uw}^\top \mathbf{X}^{12} + \mathbf{Y}^{12} M_{zy} & M_{zw}^\top \mathbf{Y}^{21} + \mathbf{Y}^{12} M_{zw} + \mathbf{Y}^{11} \end{bmatrix} > 0 \quad (9)$$

$$\begin{bmatrix} -\mathbf{Y}^{22} & -\mathbf{Y}^{22} M_{zy} & -\mathbf{Y}^{22} M_{zw} \\ -M_{zy}^\top \mathbf{Y}^{22} & -L_{uy}^\top \mathbf{X}^{12} - \mathbf{X}^{21} L_{uy} - \mathbf{X}_p^{22} + \alpha^2 \mathbf{X}_p^{11} - \alpha(L_{uy}^\top + L_{uy}) & -\mathbf{X}^{21} L_{uw} + M_{zy}^\top \mathbf{Y}^{21} + \alpha^2 \mathbf{X}_p^{11} - \alpha(L_{uy}^\top + L_{uw}) \\ -M_{zw}^\top \mathbf{Y}^{22} & -L_{uw}^\top \mathbf{X}^{12} + \mathbf{Y}^{12} M_{zy} + \alpha^2 \mathbf{X}_p^{11} - \alpha(L_{uw}^\top + L_{uy}) & M_{zw}^\top \mathbf{Y}^{21} + \mathbf{Y}^{12} M_{zw} + \mathbf{Y}^{11} + \alpha^2 \mathbf{X}_p^{11} - \alpha(L_{uw}^\top + L_{uw}) \end{bmatrix} > 0 \quad (10)$$

Box 1.

Fig. 1. A generic networked system Σ .

to the subsystem Σ_i , $i \in \mathbb{N}_N$. Consequently, w and z are structured similarly to u and y in (7).

Finally, the networked system Σ is assumed to have a set of equilibrium states $\mathcal{X} \subseteq \tilde{\mathcal{X}} \subset \mathbb{R}^n$ where $\tilde{\mathcal{X}} \triangleq \mathcal{X}_1 \times \mathcal{X}_2 \times \cdots \times \mathcal{X}_N$ and $n \triangleq \sum_{i \in \mathbb{N}_N} n_i$. Note that, based on the subsystem equilibrium points, for each $x^* \triangleq [(x_i^*)^\top]_{i \in \mathbb{N}_N}^\top \in \mathcal{X}$, there exists corresponding $u^* \triangleq [(u_i^*)^\top]_{i \in \mathbb{N}_N}^\top$ and $y^* \triangleq [(y_i^*)^\top]_{i \in \mathbb{N}_N}^\top$ that are implicit functions of x^* . Note also that, based on the networked system's equilibrium points, for each $x^* \in \mathcal{X}$, there is a unique $w^* \in \mathbb{R}^r$ that satisfies $u^* = M_{uy} y^* + M_{uw} w^*$ while both w^* and $z^* \triangleq M_{zy} y^* + M_{zw} w^*$ being implicit functions of x^* .

2.3.2. Dissipativity analysis

Inspired by Arcak (2022), the following proposition exploits the X_i -EID properties of the individual subsystems Σ_i , $i \in \mathbb{N}_N$ to formulate an LMI problem so as to analyze the $\mathbf{Y}$ -EID property of the networked system Σ , where $\mathbf{Y} = \mathbf{Y}^\top = [\mathbf{Y}^{kl}]_{k,l \in \mathbb{N}_2}$ is prespecified (e.g., based on Remark 1).

Proposition 2. The networked system Σ is $\mathbf{Y}$ -EID if there exist scalars $p_i \geq 0$, $\forall i \in \mathbb{N}_N$ such that

$$\begin{bmatrix} M_{uy} & M_{uw} \\ \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \\ M_{zy} & M_{zw} \end{bmatrix}^\top \begin{bmatrix} \mathbf{X}_p & \mathbf{0} \\ \mathbf{0} & -\mathbf{Y} \end{bmatrix} \begin{bmatrix} M_{uy} & M_{uw} \\ \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \\ M_{zy} & M_{zw} \end{bmatrix} \leq 0, \quad (8)$$

where $\mathbf{X}_p = [\mathbf{X}_p^{kl}]_{k,l \in \mathbb{N}_2}$ with $\mathbf{X}_p^{kl} \triangleq \text{diag}(p_i X_i^{kl} : i \in \mathbb{N}_N)$.

2.3.3. Topology synthesis

Inspired by Welikala et al. (2023), the next proposition formulates an LMI problem for synthesizing the interconnection matrix M (7) so as to enforce $\mathbf{Y}$ -EID property for the networked system Σ . However, as in Welikala et al. (2023), we first need to make two mild assumptions.

Assumption 1. The given $\mathbf{Y}$ -EID specification for the networked system Σ is such that $\mathbf{Y}^{22} < 0$.

Remark 2. Based on Remark 1, Assumption 1 holds if the networked system Σ is required to be: (1) L2G(γ), or (2) IF-OPF(v, ρ) with some $\rho > 0$, i.e., L_2 -stable or passive, respectively. Therefore, as it is always desirable to make the networked system Σ either L_2 -stable or passive, Assumption 1 is mild.

Assumption 2. In the networked system Σ , each subsystem Σ_i is X_i -EID with either $X_i^{11} > 0$ or $X_i^{11} < 0$, $\forall i \in \mathbb{N}_N$.

Remark 3. According to Remark 1, the above assumption directly holds if each subsystem Σ_i , $i \in \mathbb{N}_N$ is: (1) L2G(γ_i) (as $X_i^{11} = \gamma_i^2 \mathbf{I} > 0$), or (2) IF-OPF(v_i, ρ_i) with $v_i < 0$ (i.e., non-passive), or (3) IF-OPF(v_i, ρ_i) with $v_i > 0$ (i.e., passive). Therefore, Assumption 2 is also mild.

Proposition 3. Under Assumptions 1 and 2, the networked system Σ can be made $\mathbf{Y}$ -EID by synthesizing its interconnection matrix M (7) using the LMI problem:

Find: $L_{uy}, L_{uw}, M_{zy}, M_{zw}, \{p_i : i \in \mathbb{N}_N\}$

Sub. to: $p_i > 0$, $\forall i \in \mathbb{N}_N$, and

$$\begin{cases} (9), & \text{if } X_i^{11} > 0, \forall i \in \mathbb{N}_N, \\ (10) \text{ with } \alpha \in \mathbb{R}, & \text{else if } X_i^{11} < 0, \forall i \in \mathbb{N}_N, \end{cases} \quad (11)$$

(see Box 1) where $\mathbf{X}^{12} \triangleq \text{diag}((X_i^{11})^{-1} X_i^{12} : i \in \mathbb{N}_N)$, $\mathbf{X}^{21} \triangleq (\mathbf{X}^{12})^\top$ with $M_{uy} \triangleq (\mathbf{X}_p^{11})^{-1} L_{uy}$ and $M_{uw} \triangleq (\mathbf{X}_p^{11})^{-1} L_{uw}$.

Finally, we point out that the proposed synthesis technique for the interconnection matrix M given in Proposition 3 can be used even when M is partially known/fixed — as it will only reduce the number of variables in the LMI problem (11). Note also that both the proposed analysis technique given in Proposition 2 and the synthesis technique given in Proposition 3 are independent of the equilibrium points of the considered networked system.

2.4. Decentralization

To decentrally evaluate different LMI problems arising related to networked systems (e.g., for topology synthesis: (11)), in this section, we recall the concept of “network matrices” first introduced in Welikala et al. (2022a) along with some of its useful properties.

We denote a generic directed network topology as $\mathcal{G}_n = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V} \triangleq \{\Sigma_i : i \in \mathbb{N}_N\}$ is the set of nodes (subsystems), $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ is the set of edges (interconnections), and $n \in \mathbb{N}$. Corresponding to a network topology $\mathcal{G}_n$, a class of matrices named “network matrices” is defined as follows.

Definition 3 (Welikala et al., 2022a). Corresponding to a network topology $\mathcal{G}_n = (\mathcal{V}, \mathcal{E})$, any $n \times n$ block matrix $\Theta = [\Theta_{ij}]_{i,j \in \mathbb{N}_N}$ is a network matrix if: (1) Θ_{ij} consists of information specific only to the subsystems Σ_i and Σ_j , and (2) $(\Sigma_i, \Sigma_j) \notin \mathcal{E}$ and $(\Sigma_j, \Sigma_i) \notin \mathcal{E}$ implies $\Theta_{ij} = \Theta_{ji} = \mathbf{0}$, for all $i, j \in \mathbb{N}_N$.

To provide some examples, consider a network topology $\mathcal{G}_n = (\mathcal{V}, \mathcal{E})$. First, note that, any $n \times n$ block matrix $\Theta = [\Theta_{ij}]_{i,j \in \mathbb{N}_N}$ where each Θ_{ij} , $i, j \in \mathbb{N}_N$ is a coupling weight matrix specific

only to the edge $(\Sigma_i, \Sigma_j) \in \mathcal{E}$, is a network matrix. Consequently, any corresponding adjacency-like matrix is also a network matrix. Moreover, any $n \times n$ block diagonal matrix $\Theta = \text{diag}([\Theta_{ii}]_{i \in \mathbb{N}_n})$ where each Θ_{ii} , $i \in \mathbb{N}_n$ is specific only to the subsystem $\Sigma_i \in \mathcal{V}$, is a network matrix.

The following proposition, compiled from Welikala et al. (2022a), summarizes several useful properties of such network matrices.

Proposition 4 (Welikala et al., 2022a). *Given a network topology $\mathcal{G}_n$, if Θ and Φ are two corresponding network matrices, then:*

- (1) $\Theta^\top$ and $\alpha\Theta + \beta\Phi$ are network matrices for any $\alpha, \beta \in \mathbb{R}$;
- (2) $\Phi\Theta$ and $\Theta\Phi$ are network matrices whenever Φ is a block diagonal network matrix.

Moreover, if $\Psi = [\Psi^{kl}]_{k,l \in \mathbb{N}_m}$ is a block matrix where each block Ψ^{kl} , $k, l \in \mathbb{N}_m$ is a network matrix of $\mathcal{G}_n$, then:

- (3) $\text{BEW}(\Psi) \triangleq [[\Psi_{ij}^{kl}]_{k,l \in \mathbb{N}_m}]_{i,j \in \mathbb{N}_n}$ is a network matrix;
- (4) $\Psi > 0 \iff \text{BEW}(\Psi) > 0$.

The above proposition can be used to identify conditions under which certain matrices become network matrices. For example, if A and P are block network matrices and P is also block diagonal, then: $A^\top P$, PA , $A^\top P + PA$ and $\text{BEW}\left(\begin{bmatrix} P & A^\top P \\ PA & P \end{bmatrix}\right) = \begin{bmatrix} P_{ii}\Theta_{ij} & A_{ji}^\top P_{jj} \\ P_{ii}A_{ij} & P_{ii}\Theta_{ij} \end{bmatrix}_{i,j \in \mathbb{N}_n}$ are all network matrices. Moreover, it can be used to transform certain block matrix inequalities of interest into network matrix inequalities, for example, $\begin{bmatrix} P & A^\top P \\ PA & P \end{bmatrix} > 0 \iff \text{BEW}\left(\begin{bmatrix} P & A^\top P \\ PA & P \end{bmatrix}\right) > 0$.

The following proposition (inspired by Sylvester's criterion (Antsaklis & Michel, 2006), taken from Welikala et al. (2022a)) provides an approach to sequentially analyze/enforce a positive definiteness condition imposed on a symmetric block matrix.

Proposition 5 (Welikala et al., 2022a). *A symmetric $N \times N$ block matrix $W = [W_{ij}]_{i,j \in \mathbb{N}_N} > 0$ if and only if*

$$\tilde{W}_{ii} \triangleq W_{ii} - \tilde{W}_i \mathcal{D}_i \tilde{W}_i^\top > 0, \quad \forall i \in \mathbb{N}_N, \quad (12)$$

where $\tilde{W}_i \triangleq [\tilde{W}_{ij}]_{j \in \mathbb{N}_{i-1}} \triangleq W_i(\mathcal{D}_i A_i^\top)^{-1}$, $W_i \triangleq [W_{ij}]_{j \in \mathbb{N}_{i-1}}$, $\mathcal{D}_i \triangleq \text{diag}(\tilde{W}_{jj}^{-1} : j \in \mathbb{N}_{i-1})$, and A_i is the block lower-triangular matrix created from $[\tilde{W}_{kl}]_{k,l \in \mathbb{N}_{i-1}}$.

Remark 4. If W is a symmetric block network matrix corresponding to some network topology $\mathcal{G}_N$, the above proposition can be used to analyze/enforce $W > 0$ in a decentralized (as well as compositional) manner by sequentially analyzing/enforcing $\tilde{W}_{ii} > 0$ at each subsystem Σ_i , $i \in \mathbb{N}_N$. Moreover, this decentralized matrix inequality $\tilde{W}_{ii} > 0$ can be transformed into an LMI in $[W_{ij}]_{j \in \mathbb{N}_i}$ using the Schur complement theory. Therefore, such a decentralized analysis/enforcement can be executed efficiently using readily available convex optimization toolboxes (Boyd, El Ghaoui, Feron, & Balakrishnan, 1994). Additional details on Proposition 5 are omitted here but can be found in Welikala et al. (2022a), Welikala, Lin, and Antsaklis (2022b).

2.5. String stability analysis

To conclude this section, here we recall a ‘‘String Stability’’ concept that can be used to characterize how disturbances propagate over the networked systems.

Similar to before (e.g., see (6)), consider a networked system Σ comprised of N subsystems $\{\Sigma_i : i \in \mathbb{N}_N\}$, where the dynamics

of each subsystem Σ_i , $i \in \mathbb{N}_N$ are given by

$$\dot{x}_i = f_i(x_i, \{x_j\}_{j \in \mathcal{E}_i}, w_i), \quad (13)$$

where $x_i \in \mathbb{R}^{n_i}$ and $w_i \in \mathbb{R}^{r_i}$ are respectively the subsystem's state and disturbance. $\{x_j : j \in \mathcal{E}_i\}$ are the states of the neighboring subsystems of the subsystem Σ_i . Consequently, the dynamics of the networked system Σ can be written as

$$\dot{x} = f(x, w) \quad (14)$$

where $x \triangleq [x_i^\top]_{i \in \mathbb{N}_N}^\top$, $w \triangleq [w_i^\top]_{i \in \mathbb{N}_N}^\top$, $f \triangleq [f_i^\top]_{i \in \mathbb{N}_N}^\top : \mathbb{R}^n \times \mathbb{R}^r \rightarrow \mathbb{R}^n$, $n \triangleq \sum_{i \in \mathbb{N}_N} n_i$ and $r \triangleq \sum_{i \in \mathbb{N}_N} r_i$. The function f is assumed to be such that $f(x^*, 0) = 0$, $\forall x^* \in \mathcal{X} \subset \mathbb{R}^n$ ($\mathcal{X}$ denotes a set of equilibrium states), and locally Lipschitz continuous around each equilibrium point $x^* \in \mathcal{X}$.

To characterize the string stability properties of the networked system (14), we recall the definition of L_2 weak string stability initially proposed in Ploeg, Van De Wouw, and Nijmeijer (2013).

Definition 4 (Ploeg, Van De Wouw, & Nijmeijer, 2013). The networked system (14) around the equilibrium point $x^* \in \mathcal{X}$ is L_2 weakly string stable if there exist class $\mathcal{K}$ functions α_1 and α_2 such that, for any initial state $x(0) \in \mathbb{R}^n$ and $w(\cdot) \in \mathcal{L}_2^r$, the condition:

$$\|x(\cdot) - x^*\| \leq \alpha_1(\|w(\cdot)\|) + \alpha_2(\|x(0) - x^*\|) \quad (15)$$

holds for any $N \in \mathbb{N}$.

Remark 5. The above-defined L_2 weak string stability is called ‘‘weak’’ as it allows the norm $\|x(\cdot) - x^*\|$ to increase locally with respect to N (i.e., the total number of subsystems in the network) while guaranteeing the boundedness of its maximum. In particular, such a maximum is bounded independently of N , as in such a case, the condition (15) needs to hold for any $N \in \mathbb{N}$. Note also that the concept of L_2 weak string stability does not require any bounds on the external disturbances and the initial state perturbations. The ongoing research explores establishing a stronger string stability notion called *disturbance string stability* (DSS) (Mirabilio, Iovine, De Santis, Di Benedetto, & Pola, 2021) compared to L_2 weak string stability. However, due to space constraints, it is considered out of the scope of this paper.

3. Problem formulation and basic setup

3.1. Platooning problem formulation

Following Song, Welikala, Antsaklis, and Lin (2023), we consider the longitudinal dynamics of the i th vehicle Σ_i , $i \in \mathbb{N}_N \triangleq \mathbb{N}_N \cup \{0\}$ in the platoon to be

$$\Sigma_i : \begin{cases} \dot{x}_i(t) = v_i(t) + d_{xi}(t), \\ \dot{v}_i(t) = a_i(t) + d_{vi}(t), \\ \dot{a}_i(t) = f_i(v_i(t), a_i(t)) + \frac{1}{m_i \tau_i} \bar{u}_i(t) + d_{ai}(t), \end{cases} \quad (16)$$

where $f_i \triangleq f_i(v_i(t), a_i(t))$ with

$$f_i \triangleq -\frac{1}{\tau_i} \left(a_i(t) + C_{ri} + \frac{\rho A_{fi} C_{di} v_i(t)}{2m_i} (v_i(t) + 2\tau_i a_i(t)) \right).$$

Note that here we use Σ_0 to represent the leading vehicle (virtual) and $\{\Sigma_i : i \in \mathbb{N}_N\}$ to represent the following N vehicles (controllable) in the platoon. In (16), the position, velocity and acceleration of the vehicle Σ_i , $i \in \mathbb{N}_N$ are denoted by $x_i(t)$, $v_i(t)$, $a_i(t) \in \mathbb{R}$, and the corresponding disturbances are denoted by $d_{xi}(t)$, $d_{vi}(t)$, $d_{ai}(t) \in \mathbb{R}$, respectively. Further, $\bar{u}_i(t) \in \mathbb{R}$ in (16) is the control input to be designed for the follower vehicle Σ_i , $i \in \mathbb{N}_N$, and it is prespecified for the leading vehicle Σ_0 . The remaining coefficients appearing in (16) are as follows: ρ is the air density, m_i , τ_i and A_{fi} are the mass, the engine time constant

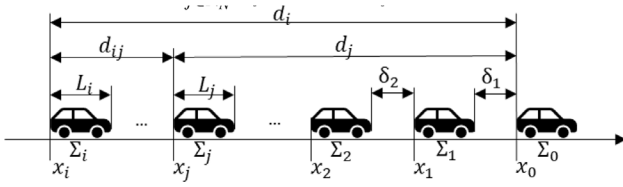


Fig. 2. Vehicle placements in the platoon.

and the effective frontal area of Σ_i , respectively, and C_{di} and C_{ri} are the coefficients of aerodynamic drag and rolling resistance of Σ_i , respectively.

Linearizing state feedback control. To facilitate our co-design process, we first introduce a linearization stage. Using the fact that $m_i, \tau_i \neq 0$ in (16), a state feedback linearizing control law for $\tilde{u}_i(t)$ can be designed as

$$\tilde{u}_i(t) = m_i \tau_i (-f_i(v_i(t), a_i(t)) + \tilde{u}_i(t)), \quad (17)$$

where $\tilde{u}_i(t)$ is the virtual control input law that needs to be designed for the follower vehicles $\Sigma_i, i \in \mathbb{N}_N$. For notational convenience, we assume that (17) is also implemented at the leading vehicle Σ_0 and $\tilde{u}_0(t)$ is prespecified. We also assume that parametric uncertainties in the linearizing control law (17) are included in the disturbance $d_{ai}(t)$ in (16). This is a mild assumption as our co-design scheme will minimize the impact of such disturbances on the platooning performance.

Consequently, under (17), the longitudinal dynamics of the vehicle $\Sigma_i, i \in \mathbb{N}_N$ take the triple integrator form:

$$\Sigma_i : \begin{cases} \dot{x}_i(t) = v_i(t) + d_{xi}(t), \\ \dot{v}_i(t) = a_i(t) + d_{vi}(t), \\ \dot{a}_i(t) = \tilde{u}_i(t) + d_{ai}(t). \end{cases} \quad (18)$$

Note that the individual vehicle dynamics (18) (or (16)) are independent of each other. However, they become dependent on each other through the used virtual feedback control law $\tilde{u}_i(t)$ - which is typically designed using the state of some corresponding error dynamics. We consider two different error dynamics formulations, as described below.

3.2. Platooning error dynamics

3.2.1. Error dynamics I

Based on the other vehicle locations $\{x_j(t) : j \in \mathbb{N}_N \setminus \{i\}\}$, here we define the *location error* of the vehicle $\Sigma_i, i \in \mathbb{N}_N$ as

$$\tilde{x}_i(t) \triangleq \sum_{j \in \mathbb{N}_N} \bar{k}_{ij} (x_i(t) - x_j(t) - d_{ij}), \quad (19)$$

where $\bar{k}_{ij} \in \mathbb{R}$ is a weighting coefficient and $d_{ij} \in \mathbb{R}$ is the desired separation between vehicles Σ_i and Σ_j . Note that $\bar{k}_{ii} \triangleq 0$ and $d_{ii} \triangleq 0$ for all $i \in \mathbb{N}_N$. Note also that, if vehicle Σ_i gets information from vehicle Σ_j (through some communication medium), then, $\bar{k}_{ij} > 0$, otherwise, $\bar{k}_{ij} = 0$. On the other hand, as shown in Fig. 2, $d_{ij} \triangleq d_i - d_j$ with $d_0 \triangleq 0$ and $d_m \triangleq \sum_{k \in \mathbb{N}_m} L_k + \delta_k$ for $m = i, j$, where L_k is the length of the vehicle Σ_k and δ_k is the desired separation between vehicles Σ_k and Σ_{k-1} . The use of location error in (19) motivates a constant spacing policy (Feng et al., 2019), which, as opposed to a velocity-dependent constant headway policy, is popular for its simplicity and resulting increased highway capacity (Jia et al., 2015; Kavathekar & Chen, 2011).

First, taking the derivative of (19) yields

$$\dot{\tilde{x}}_i(t) = \tilde{v}_i(t) + \bar{d}_{xi}(t), \quad (20)$$

where $\bar{d}_{xi}(t) \triangleq \sum_{j \in \mathbb{N}_N} \bar{k}_{ij} (d_{xi}(t) - d_{xj}(t))$ is the disturbance associated with the location error dynamics and

$$\tilde{v}_i(t) \triangleq \sum_{j \in \mathbb{N}_N} \bar{k}_{ij} (v_i(t) - v_j(t)) \quad (21)$$

is the *velocity error* of the vehicle $\Sigma_i, i \in \mathbb{N}_N$.

Next, taking the derivative of (21) yields

$$\dot{\tilde{v}}_i(t) = \sum_{j \in \mathbb{N}_N} \bar{k}_{ij} (a_i(t) - a_j(t)) + \bar{d}_{vi}(t), \quad (22)$$

where $\bar{d}_{vi}(t) \triangleq \sum_{j \in \mathbb{N}_N} \bar{k}_{ij} (d_{vi}(t) - d_{vj}(t))$ is the disturbance associated with the velocity error dynamics. However, unlike in the previous step, here we do not define the first term in (22) as the *acceleration error* due to its interconnected and sensitive nature. Instead, we define the *acceleration error* of the vehicle $\Sigma_i, i \in \mathbb{N}_N$ as

$$\tilde{a}_i(t) \triangleq (a_i(t) - a_0(t)). \quad (23)$$

Therefore, (22) can be re-stated as

$$\dot{\tilde{v}}_i(t) = \sum_{j \in \mathbb{N}_N} k_{ij} \tilde{a}_j(t) + \bar{d}_{vi}(t), \quad (24)$$

where

$$k_{ii} \triangleq \bar{k}_{i0} + \sum_{j \in \mathbb{N}_N} \bar{k}_{ij} \quad \text{and} \quad k_{ij} \triangleq -\bar{k}_{ij}, \forall j \in \mathbb{N}_N \setminus \{i\}. \quad (25)$$

It is worth noting that the coefficients $\{k_{ij} : j \in \mathbb{N}_N\}$ can uniquely determine the coefficients $\{\bar{k}_{ij} : j \in \mathbb{N}_N\}$ as $\bar{k}_{ii} \triangleq 0$.

Finally, taking the derivative of (23) yields

$$\dot{\tilde{a}}_i(t) = \tilde{u}_i(t) + \bar{d}_{ai}(t) \quad (26)$$

where $\bar{d}_{ai}(t) \triangleq -\tilde{u}_0(t) + (d_{ai}(t) - d_{a0}(t))$ is the disturbance associated with the acceleration error dynamics. Note that, here we have considered the leader's virtual control input law $\tilde{u}_0(t)$ as a disturbance because, typically, $\tilde{u}_0(t) = 0$ holds for all $t \geq 0$ except for a few time instants where the leader changes its acceleration to a different (still constant) level, e.g., see Song et al. (2022).

In all, the error dynamics of the vehicle $\Sigma_i, i \in \mathbb{N}_N$ are:

$$\begin{cases} \dot{\tilde{x}}_i(t) = \tilde{v}_i(t) + \bar{d}_{xi}(t), \\ \dot{\tilde{v}}_i(t) = \sum_{j \in \mathbb{N}_N} k_{ij} \tilde{a}_j(t) + \bar{d}_{vi}(t), \\ \dot{\tilde{a}}_i(t) = \tilde{u}_i(t) + \bar{d}_{ai}(t). \end{cases} \quad (27)$$

We select the virtual control input law $\tilde{u}_i(t)$ in the form

$$\tilde{u}_i(t) \triangleq \bar{L}_{ii} e_i(t) + u_i(t), \quad (28)$$

where $\bar{L}_{ii} \in \mathbb{R}^{1 \times 3}$ is a **local controller gain** and $e_i(t) \triangleq [\tilde{x}_i(t), \tilde{v}_i(t), \tilde{a}_i(t)]^T$. Now, by defining

$$A \triangleq \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad B \triangleq \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad (29)$$

we can restate (27) (under the choice (28)) as

$$\dot{e}_i(t) = (A + B \bar{L}_{ii}) e_i(t) + B u_i(t) + \sum_{j \in \mathbb{N}_N} \bar{K}_{ij} e_j(t) + \bar{d}_i^{(1)}(t), \quad (30)$$

where (recalling the notation: $\mathbf{e}_{ij} \triangleq \mathbf{1}_{\{i=j\}}\}$)

$$\bar{K}_{ij} \triangleq \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & k_{ij} - \mathbf{e}_{ij} \\ 0 & 0 & 0 \end{bmatrix}, \quad \bar{d}_i^{(1)}(t) \triangleq \begin{bmatrix} \bar{d}_{xi}(t) \\ \bar{d}_{vi}(t) \\ \bar{d}_{ai}(t) \end{bmatrix}. \quad (31)$$

Feedback control. To control the error dynamics (30), we select the virtual control input $u_i(t)$ as

$$u_i(t) \triangleq L_{ii} e_i(t), \quad (32)$$

where $L_{ii} \triangleq [L_{ii}^x, L_{ii}^v, L_{ii}^a] \in \mathbb{R}^{1 \times 3}$ is a **global controller gain** (besides $\{k_{ij} : j \in \mathbb{N}_N\}$, see also (28)). Now, the closed-loop error dynamics

of the vehicle Σ_i , $i \in \mathbb{N}_N$ take the form

$$\tilde{\Sigma}_i : \quad \dot{e}_i(t) = (A + B\bar{L}_{ii})e_i(t) + \eta_i(t), \quad (33)$$

where

$$\eta_i(t) \triangleq \sum_{j \in \mathbb{N}_N} K_{ij}e_j(t) + w_i(t), \quad (34)$$

with

$$K_{ij} \triangleq \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & k_{ij} - e_{ij} \\ l_{ij}^x e_{ij} & l_{ij}^v e_{ij} & l_{ij}^a e_{ij} \end{bmatrix}, \quad w_i(t) \triangleq \tilde{d}_i^{(1)}(t). \quad (35)$$

3.2.2. Error dynamics II

Compared to (19), here we define the *location error* of the vehicle Σ_i , $i \in \mathbb{N}_N$ as

$$\tilde{x}_i(t) \triangleq x_i(t) - (x_0(t) - d_{i0}), \quad (36)$$

where d_{i0} is the desired separation between vehicles Σ_i and Σ_0 (note that we use the same notation as before).

First, taking the derivative of (36) yields

$$\dot{\tilde{x}}_i(t) = \tilde{v}_i(t) + d_{xi}(t), \quad (37)$$

where

$$\tilde{v}_i(t) \triangleq v_i(t) - v_0(t) - d_{x0}(t), \quad (38)$$

is the *velocity error* of the vehicle Σ_i , $i \in \mathbb{N}_N$. Note that $d_{xi}(t)$ and $d_{x0}(t)$ are directly from (18) (or (16)).

Next, taking the derivative of (38) yields

$$\dot{\tilde{v}}_i(t) = \tilde{a}_i(t) + d_{vi}(t), \quad (39)$$

where

$$\tilde{a}_i(t) \triangleq a_i(t) - a_0(t) - d_{v0}(t) - \dot{d}_{x0}(t), \quad (40)$$

is the *acceleration error* of the vehicle Σ_i , $i \in \mathbb{N}_N$.

Finally, taking the derivative of (40) yields

$$\dot{\tilde{a}}_i(t) = \tilde{u}_i(t) + d_{ai}(t) - \ddot{u}_0(t), \quad (41)$$

where

$$\tilde{u}_0(t) \triangleq \ddot{u}_0(t) + d_{a0}(t) + \dot{d}_{v0}(t) + \ddot{d}_{x0}(t) \quad (42)$$

can be thought of as a disturbance induced by the leader's motion. Note that if we constrained the leader's motion by assuming it is noiseless and $\dot{a}_0(t) = 0$, $\forall t \in \mathbb{R}_{\geq 0}$, then, $\tilde{u}_0(t) = 0$, $\forall t \in \mathbb{R}_{\geq 0}$.

In all, the error dynamics of the vehicle Σ_i , $i \in \mathbb{N}_N$ are:

$$\begin{cases} \dot{\tilde{x}}_i(t) = \tilde{v}_i(t) + d_{xi}(t), \\ \dot{\tilde{v}}_i(t) = \tilde{a}_i(t) + d_{vi}(t), \\ \dot{\tilde{a}}_i(t) = \tilde{u}_i(t) + d_{ai}(t) - \ddot{u}_0(t). \end{cases} \quad (43)$$

Now, using the same definitions in (28) (that includes a **local controller gain** $\bar{L}_{ii}$) and (29), we can restate (43) as

$$\dot{e}_i(t) = (A + B\bar{L}_{ii})e_i(t) + Bu_i(t) + \tilde{d}_i^{(2)}(t), \quad (44)$$

where $\tilde{d}_i^{(2)}(t) \triangleq [d_{xi}(t), d_{vi}(t), d_{ai}(t) - \ddot{u}_0(t)]^\top$. Note that, unlike in (30), the error dynamics obtained in (44) are not coupled (with those of other follower vehicles). This is due to the decoupled location (36), velocity (38), and acceleration (40) errors considered in the derivation of (44).

Feedback control. To control the error dynamics (44), instead (32), here we select the virtual control input $u_i(t)$ as

$$u_i(t) = L_{ii}e_i(t) + \sum_{j \in \mathbb{N}_N \setminus \{i\}} L_{ij}(e_i(t) - e_j(t)), \quad (45)$$

where $L_{ij} \triangleq [l_{ij}^x, l_{ij}^v, l_{ij}^a] \in \mathbb{R}^{1 \times 3}$, $\forall j \in \mathbb{N}_N$ are **global controller gains**. For convenience, let us restate (45) as

$$u_i(t) = \sum_{j \in \mathbb{N}_N} \bar{K}_{ij}e_j(t) \quad (46)$$

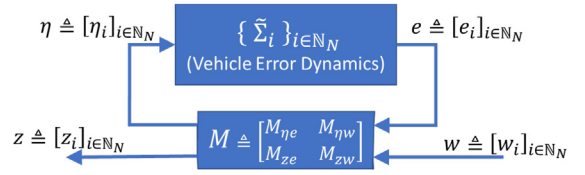


Fig. 3. Platoon error dynamics as a networked system $\tilde{\Sigma}$.

where $\bar{K}_{ij} \triangleq -L_{ij}$, $\forall j \neq i$ and $\bar{K}_{ii} \triangleq L_{ii} + \sum_{j \in \mathbb{N}_N \setminus \{i\}} L_{ij}$. Finally, by defining $[k_{ij}^x, k_{ij}^v, k_{ij}^a] \triangleq \bar{K}_{ij}$, $\forall j \in \mathbb{N}_N$, the closed-loop error dynamics of the vehicle Σ_i , $i \in \mathbb{N}_N$ can be shown to have the same form as in (33)–(34), but with

$$K_{ij} \triangleq \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ k_{ij}^x & k_{ij}^v & k_{ij}^a \end{bmatrix}, \quad w_i(t) \triangleq \tilde{d}_i^{(2)}(t), \quad (47)$$

as opposed to the choices saw in (35).

Remark 6. For a fully connected vehicular platoon, the error dynamics formulations I and II presented above have $N^2 + 6$ and $3N^2 + 3$ scalar controller gain parameters to be synthesized. This is due to their respective difference in (35) and (47) (despite sharing the same closed-loop error dynamics (33)–(34)). As a consequence, we can expect the latter approach to be less conservative while the prior approach is more computationally inexpensive to synthesize.

3.3. Networked system view

Irrespective of the used error dynamics formulation, both resulting closed-loop error dynamics share a common format (33)–(34) that can be modeled as a networked system as shown in Fig. 3 (analogous to that in Fig. 1).

In particular, here, each subsystem $\tilde{\Sigma}_i$, $i \in \mathbb{N}_N$ in the networked system (denoted $\tilde{\Sigma}$) is an LTI system $\tilde{\Sigma}_i : \eta_i \rightarrow e_i$ as given in (33)–(34). Let us collectively denote the subsystem inputs as $\eta(t) \triangleq [\eta_i^\top(t)]_{i \in \mathbb{N}_N}^\top$ and subsystem outputs as $e(t) \triangleq [e_i^\top(t)]_{i \in \mathbb{N}_N}^\top$. In addition, the overall networked system $\tilde{\Sigma}$ also include a disturbance input $w(t) \triangleq [w_i^\top(t)]_{i \in \mathbb{N}_N}^\top$ and a performance output $z(t) = [z_i^\top(t)]_{i \in \mathbb{N}_N}^\top$. These four components/ports are interconnected according to the relationship:

$$\begin{bmatrix} \eta \\ z \end{bmatrix} = \begin{bmatrix} M_{\eta e} & M_{\eta w} \\ M_{ze} & M_{zw} \end{bmatrix} \begin{bmatrix} e \\ w \end{bmatrix} \equiv M \begin{bmatrix} e \\ w \end{bmatrix}, \quad (48)$$

where M is the interconnection matrix, $M_{\eta e} \triangleq [K_{ij}]_{i,j \in \mathbb{N}_N}$ (from (35) or (47)), $M_{\eta w} \triangleq \mathbf{I}$, $M_{ze} \triangleq \mathbf{I}$, and $M_{zw} \triangleq \mathbf{0}$.

Based on this networked system view of the closed-loop error dynamics of the platoon, we can now apply Proposition 3 to synthesize a suitable $M_{\eta e} \triangleq [K_{ij}]_{i,j \in \mathbb{N}_N}$ while constraining each K_{ij} , $i, j \in \mathbb{N}_N$ to be of the form (35) or (47) (depending on the preferred error dynamics formulation). Consequently, it can be used to obtain the corresponding controller gains required to evaluate the virtual control inputs (32) or (45). Finally, we conclude this section by pointing out that synthesizing $M_{\eta e}$ will reveal not only the desired individual controllers but also a preferable communication topology for the platoon.

4. Centralized design of the platoon

In this section, we discuss a centralized LMI-based technique to design the platoon, i.e., to obtain the local and global controller gains and the communication topology. Note that even though this centralized design process is not compositional, it

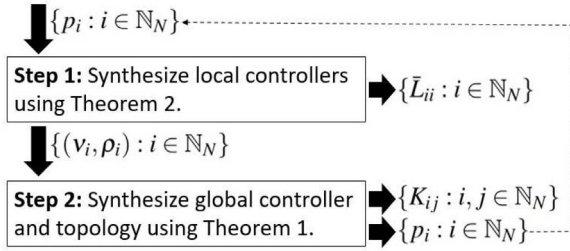


Fig. 4. The overall centralized co-design process.

lays the foundation for creating its decentralized version, which is compositional (presented in Section 5).

The overall centralized co-design process for designing local controllers, global controllers, and communication topology can be broken down into two steps, as shown in Fig. 4. The first step involves synthesizing a local controller at each vehicle to establish favorable passivity indices for its platooning error dynamics. The second step involves synthesizing the global controllers and the communication topology to jointly optimize the disturbance robustness and the total communication cost of the closed-loop vehicular platoon. These two steps are achieved by solving the LMI problems formulated in Theorems 1 and 2, and their details are provided in Sections 4.1 and 4.3 (see also Remark 9).

Since we view the overall closed-loop error dynamics of the platoon as a networked system $\tilde{\Sigma}$ (see Fig. 3) and intend to apply Proposition 3 for the global controller and topology design, we require the dissipativity properties of the individual subsystems, i.e., of the individual vehicle closed-loop error dynamics $\tilde{\Sigma}_i, i \in \mathbb{N}_N$ (33). However, based on (33) and Corollary 1, it is clear that such subsystem dissipativity properties will be dependent upon the used local controllers $\tilde{L}_{ii}, i \in \mathbb{N}_N$ (28). Consequently, designing local and global controllers (along with the topology) are two highly coupled problems.

However, solving such two coupled problems together as a single combined problem is not practical as such a combined problem will be a larger, non-decentralizable, and non-linear matrix inequality problem – as opposed to being a smaller, decentralizable, and LMI problem (e.g., see (5) and (8)).

On the other hand, we also cannot solve these two coupled problems as two entirely independent/decoupled problems, as subsystem dissipativity properties enforced by an independent local controller design, may not be sufficient/suitable for obtaining a feasible global controller design for the networked system. Conversely, subsystem dissipativity properties required for obtaining a feasible global controller design for the networked system may not be achievable by an independent local controller design. Moreover, the optimality of the overall design is also a concern if these two coupled problems are to be treated as two entirely independent/decoupled problems.

To address this challenge, in this paper, we consider these two problems as two “loosely coupled” problems. In particular, we include several additional parameters and conditions in the local controller design to ensure that the resulting subsystem dissipativity properties are feasible and effective for use in the global controller design. Moreover, the proposed loosely coupled approach to designing local and global controllers is also decentralizable due to its emphasis on strengthening the local controller design at subsystems.

This section is arranged as follows. We first present the global controller design problem for the networked system. Next, we identify several necessary conditions on the subsystem dissipativity properties to achieve a feasible and effective global controller design. Subsequently, we present the (strengthened) local

controller design problem that enforces the identified necessary conditions. Finally, we summarize the overall local and global controller design process.

4.1. Global controller and topology synthesis

Assume each subsystem $\tilde{\Sigma}_i, i \in \mathbb{N}_N$ (33) to be X_i -EID with

$$X_i = \begin{bmatrix} X_i^{11} & X_i^{12} \\ X_i^{21} & X_i^{22} \end{bmatrix} \triangleq \begin{bmatrix} -v_i \mathbf{I} & \frac{1}{2} \mathbf{I} \\ \frac{1}{2} \mathbf{I} & -\rho_i \mathbf{I} \end{bmatrix}, \quad (49)$$

i.e., IF-OPF(v_i, ρ_i) (see Remark 1), where passivity indices v_i and ρ_i are fixed and known (we will relax this assumption later on). These subsystem EID properties will be used to apply Proposition 3 to synthesize the interconnection matrix block $M_{\eta e} = [K_{ij}]_{i,j \in \mathbb{N}_N}$ (48). Upon the synthesis of matrices $[K_{ij}]_{i,j \in \mathbb{N}_N}$, we can uniquely determine the controller gains $\{\bar{k}_{ij} : i, j \in \mathbb{N}_N\}$ (19) or $\{L_{ij} : i, j \in \mathbb{N}_N\}$ (45), depending on the used error dynamics formulation. Moreover, it will reveal the desired communication topology for the platoon.

As the main specification for this controller and topology synthesis, we require the overall platoon error dynamics (i.e., the networked system $\tilde{\Sigma}$ in Fig. 3) to be finite-gain L_2 -stable with some L_2 -gain γ from its disturbance input $w(t)$ to performance output $z(t)$. Particularly, letting $\tilde{\gamma} \triangleq \gamma^2$ we require $0 < \tilde{\gamma} < \bar{\gamma}$ where $\bar{\gamma} \in \mathbb{R}_{>0}$ is a pre-specified constant. Based on Remark 1, this specification is akin to requiring the networked system $\tilde{\Sigma}$ to be $\mathbf{Y}$ -EID with

$$\mathbf{Y} = \begin{bmatrix} \gamma^2 \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{I} \end{bmatrix} = \begin{bmatrix} \tilde{\gamma} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{I} \end{bmatrix} \quad (50)$$

and $0 < \tilde{\gamma} < \bar{\gamma}$. Note also that we can further minimize $\tilde{\gamma}$ below $\bar{\gamma}$ by including the $\tilde{\gamma}$ term in the objective function to be optimized in this synthesis.

Apart from the said L_2 -gain γ related term $\tilde{\gamma}$, in this objective function, we also propose to include a term analogous to $\sum_{i,j \in \mathbb{N}_N} c_{ij} \|K_{ij}\|_1$ where $C = [c_{ij}]_{i,j \in \mathbb{N}_N} \in \mathbb{R}^{N \times N}$ is a prespecified cost matrix corresponding to the inter-vehicle communication topology in the platoon. For example, suppose $N = 4$ and the platoon requires a uniform separation among the vehicles. Then, to penalize inter-vehicle communication links based on their lengths, one can use the cost matrix $C \triangleq [|i - j|]_{i,j \in \mathbb{N}_4}$. Note that we propose to use an objective function containing 1-norms of the design variables (i.e., $\{\|K_{ij}\|_1 : i, j \in \mathbb{N}_N\}$ as they are well-known to induce sparsity in the solution (Sun, Welikala, & Cassandras, 2020; Tibshirani, 1996), thus leading to a sparse communication topology that is easy to implement and maintain.

Considering these concerns, the following theorem formulates the centralized controller and topology synthesis problem as an LMI problem that can be solved using standard convex optimization toolboxes (Boyd et al., 1994; Lofberg, 2004).

Theorem 1. The closed-loop error dynamics of the platoon (shown in Fig. 3) can be made finite-gain L_2 -stable with some L_2 -gain γ (where $\tilde{\gamma} \triangleq \gamma^2 < \bar{\gamma}$) from disturbance input $w(t)$ to performance output $z(t)$, by using the interconnection matrix block $M_{\eta e} = [K_{ij}]_{i,j \in \mathbb{N}_N}$ (48) synthesized via solving the LMI problem:

$$\min_{Q, \tilde{\gamma}, \{p_i : i \in \mathbb{N}_N\}} \sum_{i,j \in \mathbb{N}_N} c_{ij} \|Q_{ij}\|_1 + c_0 \tilde{\gamma}, \quad (51a)$$

$$\text{Sub. to: } p_i > 0, \forall i \in \mathbb{N}_N, \quad 0 < \tilde{\gamma} < \bar{\gamma} \quad \text{and} \quad \begin{bmatrix} \mathbf{X}_p^{11} & \mathbf{0} & \mathbf{0} & \mathbf{X}_p^{11} \\ \mathbf{0} & \mathbf{I} & \mathbf{I} & \mathbf{0} \\ Q^T & \mathbf{I} & -Q^T \mathbf{X}^{12} - \mathbf{X}^{21} Q - \mathbf{X}_p^{22} & -\mathbf{X}^{21} \mathbf{X}_p^{11} \\ \mathbf{X}_p^{11} & \mathbf{0} & -\mathbf{X}_p^{11} \mathbf{X}^{12} & \tilde{\gamma} \mathbf{I} \end{bmatrix} > 0, \quad (51b)$$

where $c_0 > 0$ is a prespecified constant, $Q \triangleq [Q_{ij}]_{i,j \in \mathbb{N}_N}$ shares the same structure as $M_{\eta e}$, $\mathbf{X}^{12} \triangleq \text{diag}(-\frac{1}{2v_i} \mathbf{I} : i \in \mathbb{N}_N)$, $\mathbf{X}^{21} \triangleq (\mathbf{X}^{12})^T$,

$\mathbf{X}_p^{11} \triangleq \text{diag}(-p_i v_i \mathbf{I} : i \in \mathbb{N}_N)$, $\mathbf{X}_p^{22} \triangleq \text{diag}(-p_i \rho_i \mathbf{I} : i \in \mathbb{N}_N)$, and $M_{\eta e} \triangleq (\mathbf{X}_p^{11})^{-1} \mathbf{Q}$.

Proof. The proof follows from applying the subsystem dissipativity properties assumed in (49) and the networked system dissipativity properties considered in (50) in the interconnection topology synthesis result stated in Proposition 3. ■

Note that the interconnection matrix block $M_{\eta e} = [K_{ij}]_{i,j \in \mathbb{N}_N}$ synthesized in Theorem 1 provides both the vehicle (global) controller gains and the inter-vehicle communication topology for the platoon that optimally attenuates the disturbance effects on the performance while also minimizing a measure of total communication cost. This is clear from the optimized joint objective function in (51). Simply, the co-design process outlined in Theorem 1 leads not only to improving the disturbance robustness but also to minimizing the total communication cost of the vehicular platoon.

Note also that even though the information exchanged in a single communication link is small, such links have to be properly established, frequently used, and regularly maintained – which may be costly. Consequently, the proposed control and communication co-design process reveals which communication links are the most cost-effective for disturbance robustness (and for string stability, see Theorem 4 in the sequel) of the vehicular platoon. Hence the proposed co-design framework provides insights into cost-effective communication topologies for vehicular platoons.

For completeness, here we provide the direct relationship between the synthesized interconnection matrix block $[K_{ij}]_{i,j \in \mathbb{N}_N}$ in Theorem 1 and the individual vehicle (global) controller gains required in (19) or (45). In particular, under the error dynamics formulations I and II, the off-diagonal elements of $[K_{ij}]_{i,j \in \mathbb{N}_N}$ are, respectively,

$$K_{ij} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\tilde{k}_{ij} \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad K_{ij} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -\tilde{l}_{ij}^x & -\tilde{l}_{ij}^v & -\tilde{l}_{ij}^a \end{bmatrix},$$

for all $i \in \mathbb{N}_N, j \in \mathbb{N}_N \setminus \{i\}$. The diagonal elements are

$$K_{ii} = K_{i0} - \sum_{j \in \mathbb{N}_N \setminus \{i\}} K_{ij}, \quad (52)$$

for all $i \in \mathbb{N}_N$, where each K_{i0} , $i \in \mathbb{N}_N$, under the error dynamics formulations I and II, are, respectively,

$$K_{i0} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \tilde{k}_{i0} - 1 \\ \tilde{l}_{ii}^x & \tilde{l}_{ii}^v & \tilde{l}_{ii}^a \end{bmatrix} \quad \text{and} \quad K_{i0} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \tilde{l}_{ii}^x & \tilde{l}_{ii}^v & \tilde{l}_{ii}^a \end{bmatrix}.$$

4.2. Necessary conditions on subsystem passivity indices

Based on (51b) in Theorem 1 (particularly, see the terms involving $\mathbf{X}_p^{11}$, $\mathbf{X}_p^{22}$, $\mathbf{X}^{12}$ and $\mathbf{X}^{21}$), it is clear that both the feasibility and the effectiveness of the global controller design hinges on the used subsystem passivity indices $\{v_i, \rho_i : i \in \mathbb{N}_N\}$ (49) assumed for the subsystems $\{\tilde{\Sigma}_i : i \in \mathbb{N}_N\}$ (33).

However, based on Corollary 1, local controllers $\{\tilde{L}_{ii} : i \in \mathbb{N}_N\}$ (28) can be designed for the subsystems $\{\tilde{\Sigma}_i : i \in \mathbb{N}_N\}$ (33) to achieve a certain set of subsystem passivity indices (e.g., the passivity indices that maximizes $v_i + \rho_i, i \in \mathbb{N}_N$), independently from the global controller design (51).

As mentioned earlier, executing the local controller designs independently can lead to an infeasible and/or ineffective global controller design. Therefore, local controller designs must account for specific conditions required in global controller design (51). A few such conditions have been identified in the following lemma.

Lemma 1. For the main LMI condition (51b) in Theorem 1 to hold, it is necessary that, at each subsystem $\tilde{\Sigma}_i, i \in \mathbb{N}_N$ (33), for some scalar parameter $p_i > 0$, the passivity indices v_i and ρ_i (49) (let $\tilde{\rho}_i \triangleq \frac{1}{\rho_i}$) are such that the LMI problem:

$$\text{Find: } v_i, \tilde{\rho}_i, \tilde{\gamma}_i, \quad (53a)$$

$$\text{Sub. to: } -\frac{\tilde{\gamma}_i}{p_i} < v_i < 0, \quad (53b)$$

$$0 < \tilde{\rho}_i < p_i, \quad (53c)$$

$$0 < \tilde{\rho}_i < \frac{4\tilde{\gamma}_i}{p_i}, \quad (53d)$$

is feasible.

Proof. Note that (51b) holds if there exists $\tilde{\Gamma} \triangleq \text{diag}(\tilde{\gamma}_i \mathbf{I} : i \in \mathbb{N}_N)$ such that $0 < \tilde{\Gamma} < \tilde{\gamma} \mathbf{I}$ (equivalently, if there exists some scalars $\tilde{\gamma}_i, \forall i \in \mathbb{N}_N$ such that $0 < \tilde{\gamma}_i < \tilde{\gamma}, \forall i \in \mathbb{N}_N$) and

$$\begin{bmatrix} \mathbf{X}_p^{11} & \mathbf{0} & \mathbf{Q} & \mathbf{X}_p^{11} \\ \mathbf{0} & \mathbf{I} & \mathbf{I} & \mathbf{0} \\ \mathbf{Q}^\top & \mathbf{I} & -\mathbf{Q}^\top \mathbf{X}^{12} - \mathbf{X}^{21} \mathbf{Q} - \mathbf{X}_p^{22} & -\mathbf{X}^{21} \mathbf{X}_p^{11} \\ \mathbf{X}_p^{11} & \mathbf{0} & -\mathbf{X}_p^{11} \mathbf{X}^{12} & \tilde{\Gamma} \end{bmatrix} > 0. \quad (54)$$

Now, for the LMI condition (54) to hold, the following set of necessary conditions can be identified (by considering some specific sub(block)matrices of the block matrix in (54)):

$$\mathbf{X}_p^{11} > 0 \quad (55a)$$

$$-\mathbf{Q}^\top \mathbf{X}^{12} - \mathbf{X}^{21} \mathbf{Q} - \mathbf{X}_p^{22} > 0 \quad (55b)$$

$$\begin{bmatrix} \mathbf{X}_p^{11} & \mathbf{X}_p^{11} \\ \mathbf{X}_p^{11} & \tilde{\Gamma} \end{bmatrix} > 0, \quad (55c)$$

$$\begin{bmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{I} & -\mathbf{Q}^\top \mathbf{X}^{12} - \mathbf{X}^{21} \mathbf{Q} - \mathbf{X}_p^{22} \end{bmatrix} > 0, \quad (55d)$$

$$\begin{bmatrix} -\mathbf{Q}^\top \mathbf{X}^{12} - \mathbf{X}^{21} \mathbf{Q} - \mathbf{X}_p^{22} & -\mathbf{X}^{21} \mathbf{X}_p^{11} \\ -\mathbf{X}_p^{11} \mathbf{X}^{12} & \tilde{\Gamma} \end{bmatrix} > 0, \quad (55e)$$

A necessary condition for (55a) can be obtained as

$$(55a) \iff -p_i v_i > 0 \iff v_i < 0, \forall i \in \mathbb{N}_N. \quad (56)$$

Considering the diagonal elements of the matrix in (55b) and then using structure of the matrix $\mathbf{Q}$ (particularly, using the fact that each $\mathbf{Q}_{ii}$ block has zeros in its diagonal, for any $i \in \mathbb{N}_N$), a necessary condition for (55b) can be obtained as

$$(55b) \implies -\mathbf{Q}_{ii}^\top \left(-\frac{1}{2v_i} \mathbf{I}\right) - \left(-\frac{1}{2v_i} \mathbf{I}\right) \mathbf{Q}_{ii} - (-p_i \rho_i \mathbf{I}) > 0 \\ \implies p_i \rho_i > 0 \iff \rho_i > 0, \forall i \in \mathbb{N}_N. \quad (57)$$

To obtain the necessary conditions for (55c)–(55e), apart from the above two techniques, we also need to use the result in Proposition 4 (Part 4). Consequently, we can obtain:

$$(55c) \iff \begin{bmatrix} -p_i v_i & -p_i v_i \\ -p_i v_i & \tilde{\gamma}_i \end{bmatrix} > 0 \iff -p_i v_i (\tilde{\gamma}_i + p_i v_i) > 0 \\ \iff v_i > -\frac{\tilde{\gamma}_i}{p_i}, \forall i \in \mathbb{N}_N; \quad (58)$$

$$(55d) \implies \begin{bmatrix} 1 & 1 \\ 1 & p_i \rho_i \end{bmatrix} > 0 \iff \rho_i > \frac{1}{p_i}, \forall i \in \mathbb{N}_N; \quad (59)$$

$$(55e) \implies \begin{bmatrix} p_i \rho_i & -\frac{1}{2} p_i \\ -\frac{1}{2} p_i & \tilde{\gamma}_i \end{bmatrix} > 0 \iff p_i (\rho_i \tilde{\gamma}_i - \frac{1}{4} p_i) > 0 \\ \iff \rho_i > \frac{p_i}{4\tilde{\gamma}_i}, \forall i \in \mathbb{N}_N. \quad (60)$$

Finally, by: (1) compiling the necessary conditions found in (56)–(60), (2) using the change of variables $\tilde{\rho}_i \triangleq \frac{1}{\rho_i}$, $\forall i \in \mathbb{N}_N$, and (3) considering each p_i , $i \in \mathbb{N}_N$ as a predefined positive scalar parameter, we can obtain the LMI problems (each in variables v_i , $\tilde{\rho}_i$, $\tilde{\gamma}_i$, for any $i \in \mathbb{N}_N$) given in (53). ■

Regarding the predefined parameters $\{p_i : i \in \mathbb{N}_N\}$ and the LMI variables $\{\tilde{\gamma}_i : i \in \mathbb{N}_N\}$ seen in the LMI problems (53) (and also their respective counterparts: the LMI variables $\{p_i : i \in \mathbb{N}_N\}$ and $\tilde{\gamma}$ seen in the LMI problem (51)), the following two respective remarks can be made.

Remark 7. In the LMI problems (53), the reason for considering the parameters $\{p_i : i \in \mathbb{N}_N\}$ as “predefined” (as opposed to LMI variables) is that they appear non-linearly with the other LMI variables $\{v_i, \tilde{\rho}_i, \tilde{\gamma}_i : i \in \mathbb{N}_N\}$. Note, however, that, these predefined parameters $\{p_i : i \in \mathbb{N}_N\}$ are not required to be equal to the eventual values of the LMI variables $\{p_i : i \in \mathbb{N}_N\}$ seen in the LMI problem (51). This is because, if for some set of predefined parameters $\{p_i : i \in \mathbb{N}_N\}$, the LMI problems (53) are feasible, then it in itself can be thought of as a necessary condition for the LMI problem (51). Note also that these predefined parameters $\{p_i : i \in \mathbb{N}_N\}$ also enable the optimization of the overall local and global controller design process. Regardless, in practice, particularly for homogeneous networks, selecting $p_i = \frac{1}{N}$, $\forall i \in \mathbb{N}_N$ in the LMI problems (53) has rendered reasonable local and global controller designs.

Remark 8. Similar to the set $\{p_i : i \in \mathbb{N}_N\}$, one can also consider the set $\{\tilde{\gamma}_i : i \in \mathbb{N}_N\}$ in the LMI problems (53) as a set of “predefined” parameters. However, it is not necessary as $\{\tilde{\gamma}_i : i \in \mathbb{N}_N\}$ appear linearly with the other LMI variables $\{v_i, \tilde{\rho}_i : i \in \mathbb{N}_N\}$ in (53). Further, under the same arguments made in Remark 7, note that these LMI variables $\{\tilde{\gamma}_i : i \in \mathbb{N}_N\}$ are not required to relate (in any way) to the LMI variable $\tilde{\gamma}$ seen in the LMI problem (51). Note also that, minimizing these LMI variables $\tilde{\gamma}_i$ in the LMI problems (53) lead to higher passivity indices $\{v_i, \rho_i : i \in \mathbb{N}_N\}$ from (53). And consequently, it can lead to a lower/better $\tilde{\gamma}$ value in the LMI problem (51). Therefore, these LMI variables $\{\tilde{\gamma}_i : i \in \mathbb{N}_N\}$ also enable the optimization of the overall local and global controller design process.

4.3. Local controller synthesis

Having established the fundamental result in Corollary 1, the subsystem (vehicle error dynamics) models in (33), and the necessary conditions on subsystem passivity indices in Lemma 1, we are now ready to present the local controller synthesis problem as an LMI problem formally in a theorem.

Theorem 2. At each vehicle Σ_i , $i \in \mathbb{N}_N$, for some prespecified scalar parameter $p_i > 0$, to make the corresponding closed-loop error dynamics (subsystem) $\tilde{\Sigma}_i$ (33) IF-OPF(v_i, ρ_i) (as required in (49)) while also satisfying the necessary conditions identified in Lemma 1, the local controller gains $\tilde{L}_{ii}$ (28) should be synthesized via solving the LMI problem:

Find: $\tilde{L}_{ii}$, P_i , v_i , $\tilde{\rho}_i$, $\tilde{\gamma}_i$,

Sub. to: $P_i > 0$,

$$\begin{bmatrix} \tilde{\rho}_i \mathbf{I} & P_i & \mathbf{0} \\ P_i & -\mathcal{H}(AP_i + B\tilde{L}_{ii}) & -I + \frac{1}{2}P_i \\ \mathbf{0} & -I + \frac{1}{2}P_i & -v_i \mathbf{I} \end{bmatrix} > 0, \quad (61)$$

$$-\frac{\tilde{\gamma}_i}{p_i} < v_i < 0,$$

$$0 < \tilde{\rho}_i < \min \left\{ p_i, \frac{4\tilde{\gamma}_i}{p_i} \right\}$$

where $\rho_i \triangleq \frac{1}{\tilde{\rho}_i}$ and $\tilde{L}_{ii} \triangleq \tilde{L}_{ii}P_i^{-1}$.

Proof. The proof follows from: (1) considering the vehicle error dynamics (subsystem) model (33) and then applying the LMI-based controller synthesis technique given in Corollary 1 to enforce the subsystem passivity indices required in (49), and then (2) enforcing the necessary conditions on subsystem passivity indices identified in Lemma 1 to ensure a feasible and effective global controller design at (51). ■

Remark 9. Apart from executing the two steps of the overall co-design process (see Fig. 4) just once, they can also be implemented iteratively because Step 2 may reveal a more suitable set of parameters $\{p_i : i \in \mathbb{N}_N\}$ to be used in a Step 1 in a following iteration. The effectiveness and the convergence of such an iterative co-design scheme is the subject of ongoing research and is considered beyond the scope of this paper. On the other hand, one can also use a brute force or non-linear optimization approach to find a candidate set of parameters $\{p_i : i \in \mathbb{N}_N\}$ that renders a lower objective function value at Step 2.

Remark 10. According to (33), subsystems $\{\tilde{\Sigma}_i : i \in \mathbb{N}_N\}$ are linear and homogeneous. This is mainly a consequence of the used feedback linearization step (17). If needed, one can promote/retain heterogeneity in (33) by modifying (17) to exclude some natural and helpful terms (e.g., $-\frac{1}{\tau_i}a_i$) from being canceled in the original vehicle dynamics (16). Similarly, we can also retain/promote some non-linearities in (33) (from (16)), particularly if they are helpful and lead to stronger dissipativity properties than those can be established via Theorem 2. In essence, such flexibilities highlight the impact of the proposed dissipativity-based approach.

5. Decentralize design of the platoon

In this section, we first show how the centralized controller and topology co-design problems formulated in Theorems 1 and 2 can be executed in a decentralized and compositional manner over the platoon. Subsequently, exploiting the said compositionality features, we detail how scenarios like merging and splitting in vehicular platoons can be handled efficiently (as opposed to directly using the centralized co-design). Finally, we discuss the analysis and enforcement of the string stability properties of the resulting platoons.

5.1. Decentralized co-design of controllers and topology

To decentrally implement the LMI problem (51), we cannot directly apply the network matrices-based decentralization technique proposed in Welikala et al. (2022a). This is due to the dependence of the diagonal blocks in $M_{\eta e} = [K_{ij}]_{i,j \in \mathbb{N}_N}$ (synthesized in (51)) on its off-diagonal blocks (52), and the presence of the global constraint $\tilde{\gamma} < \bar{\gamma}$. To overcome these two challenges, we propose to exploit the term K_{i0} included in each diagonal term K_{ii} in $M_{\eta e}$, and the concept of diagonal dominance, respectively. The following theorem provides the proposed decentralized implementation of the LMI problems (51) and (61) that can be used to decentrally and compositionally synthesize the controllers and the topology for the considered vehicular platoon.

Theorem 3. The closed-loop error dynamics of the platoon (shown in Fig. 3) can be made finite-gain L_2 -stable with some L_2 -gain γ (where $\tilde{\gamma} \triangleq \gamma^2 < \bar{\gamma}$ from disturbance input $w(t)$ to performance output $z(t)$), if at each vehicle Σ_i , $i \in \mathbb{N}_N$: (1) the local controller

gains $\tilde{L}_{ii}$ (33) are designed using the local LMI problem (61), (2) the interconnection/global controller gain blocks $\{K^i\}$ are designed using the local LMI problem:

$$\min_{\{Q^i\}, \hat{\gamma}_i, p_i} \sum_{j \in \mathbb{N}_{i-1}} c_{ij} \|Q_{ij}\|_1 + c_{ji} \|Q_{ji}\|_1 + c_{0i} \hat{\gamma}_i + c_i |\hat{\gamma}_i - \tilde{\gamma}_i| \quad (62)$$

Sub. to: $p_i > 0$, $\hat{\gamma}_i < \tilde{\gamma}$, $\tilde{W}_{ii} > 0$,

where $\tilde{\gamma}_i$ is from (61) (obtained in Step 1), $\tilde{W}_{ii}$ is from (12) when enforcing $W = [W_{ij}]_{i,j \in \mathbb{N}_N} > 0$ with

$$W_{ij} \triangleq \begin{bmatrix} e_{ij} V_p^{ii} & \mathbf{0} & Q_{ij} & e_{ij} V_p^{ii} \\ \mathbf{0} & e_{ij} \mathbf{I} & e_{ij} \mathbf{I} & \mathbf{0} \\ Q_{ji}^\top & e_{ji} \mathbf{I} & -Q_{ji}^\top S_{jj} - S_{ji} Q_{ij} - e_{ji} R_p^{ii} & -e_{ji} S_{ji} V_p^{ii} \\ e_{ij} V_p^{ii} & \mathbf{0} & -e_{ij} V_p^{ii} S_{jj} & \hat{\gamma}_i e_{ij} \mathbf{I} \end{bmatrix}, \quad (63)$$

$V_p^{ii} \triangleq -p_i v_i \mathbf{I}$, $R_p^{ii} \triangleq -p_i \rho_i \mathbf{I}$, $S_{ii} \triangleq -\frac{1}{2v_i} \mathbf{I}$ and blocks $\{K^i\}$ are determined by $K_{ij} = (V_p^{ii})^{-1} Q_{ij}$, and (3) the update:

(using the obtained blocks $\{K_{ji} : j \in \mathbb{N}_{i-1}\}$ in Step 2) is requested at each prior neighbors $\Sigma_j, j \in \mathbb{N}_{i-1} \cap \mathcal{F}_i$.

Proof. According to Proposition 5, enforcing $\tilde{W}_{ii} > 0$ at each vehicle $\Sigma_i, i \in \mathbb{N}_N$ is equivalent to enforcing $W = [W_{ij}]_{i,j \in \mathbb{N}_N} > 0$ for the entire platoon Σ . It is easy to see that this $W = \text{BEW}(\tilde{W})$, where $\tilde{W}$ is (also using some notations in Theorem 1):

$$\tilde{W} \triangleq \begin{bmatrix} \mathbf{X}_p^{11} & \mathbf{0} & Q & \mathbf{X}_p^{11} \\ \mathbf{0} & \mathbf{I} & \mathbf{I} & \mathbf{0} \\ Q^\top & \mathbf{I} & -Q^\top \mathbf{X}^{12} - \mathbf{X}^{21} Q - \mathbf{X}_p^{22} & -\mathbf{X}^{21} \mathbf{X}_p^{11} \\ \mathbf{X}_p^{11} & \mathbf{0} & -\mathbf{X}_p^{11} \mathbf{X}^{12} & \Gamma \end{bmatrix} \quad (64)$$

with $\Gamma \triangleq \text{diag}([\hat{\gamma}_i \mathbf{I}]_{i \in \mathbb{N}_N})$. Let us define $\tilde{\gamma} \triangleq \max_{i \in \mathbb{N}_N} \hat{\gamma}_i$, which satisfies $\tilde{\gamma} < \tilde{\gamma}$ due to local enforcements of $\hat{\gamma}_i < \tilde{\gamma}$ at each vehicle $\Sigma_i, i \in \mathbb{N}_N$, and $\Gamma < \tilde{\gamma} \mathbf{I}$ due to the diagonal dominance concept. Since $\{\tilde{W}_{ii} > 0 : \forall i \in \mathbb{N}_N\} \iff W = \text{BEW}(\tilde{W}) > 0 \iff \tilde{W} > 0 \implies (51b)$ (respectively from Proposition 5, Proposition 4 and the diagonal dominance concept), and $\{\hat{\gamma}_i < \tilde{\gamma} : \forall i \in \mathbb{N}_N\} \implies \tilde{\gamma} < \tilde{\gamma}$, it is clear that any solution that satisfies the decentralized set of LMI conditions in (62) will also satisfy the centralized set of LMI conditions in (51).

However, there is one remaining concern due to the special dependence explained in (52). Note that the proposed decentralized process (62) derives the interconnection/controller gain blocks iteratively, i.e., at each vehicle/iteration $\Sigma_i, i \in \mathbb{N}_N$, the set of blocks $\{K^i\} \triangleq \{K_{ii}\} \cup \{K_{ij} : j \in \mathbb{N}_{i-1}\} \cup \{K_{ji} : j \in \mathbb{N}_{i-1}\}$ is derived. In such an iteration (say at vehicle $\Sigma_i, i \in \mathbb{N}_N$), due to (52), each derived matrix $K_{ji}, j \in \mathbb{N}_{i-1}$ will affect the matrix K_{ij} derived previously at the prior neighboring vehicle Σ_j (of vehicle Σ_i). Explicitly,

$$K_{ij}^{\text{New}} = \left(K_{j0} - \sum_{l < i, l \neq j} K_{jl} \right) - K_{ji} = K_{ij}^{\text{Old}} - K_{ji}. \quad (65)$$

Since the proposed decentralization (i.e., the application of Proposition 5) is valid only if W is a network matrix, we require both Q , and by extension, $[K_{ij}]_{i,j \in \mathbb{N}_N} = [(V_p^{ii})^{-1} Q_{ij}]_{i,j \in \mathbb{N}_N}$, to be network matrices (see Proposition 4). The latter requirement holds only if we constrain $K_{ij}^{\text{New}} = K_{ij}^{\text{Old}}$. For this, we require the updates given in (63) - which completes the proof. ■

Remark 11. The design of local controllers given in Theorem 2, due to its decentralized form, seamlessly fits in with the overall decentralized solution given in Theorem 3. Consequently, the local controller design (under Step 1 of Theorem 3) enforces a

favorable set of local passivity indices for the subsequent decentralized global controller design (under Step 2 of Theorem 3). Note, however, that the objective function of this decentralized global controller design (62) now includes a term $|\hat{\gamma}_i - \tilde{\gamma}_i|$ that penalizes the mismatch between the local and global controller designs. This is to prevent such a mismatch from growing over the decentralized iterations in Theorem 3.

Remark 12. Even though we can use Theorem 3 to decentrally synthesize the controllers and topology so that a global constraint like $\gamma^2 \leq \tilde{\gamma}$ holds for the platoon, its optimality is not guaranteed when compared to the centrally synthesized controllers and topology given by Theorem 1. However, the LMI conditions in the decentralized implementation (62) are more flexible than those in the centralized implementation (51) due to the involved additional design variables in (62). Regardless, the decentrally synthesized controllers and topology can be more conservative than their centralized counterparts due to the update steps (63). In the ongoing research, we attempt to manipulate the cost function coefficients $\{c_{ij} : i, j \in \mathbb{N}_N\} \cup \{(c_{0i}, c_i) : i \in \mathbb{N}_N\}$ in (62) to mitigate such optimality and conservativeness concerns.

Remark 13. One advantage of the proposed decentralized design (62) is that it can encode custom string stability conditions. For example, let us consider a custom string stability condition for the platoon error dynamics:

$$\sup_{\{w_j(\cdot)\}_{j \in \mathbb{N}_i}} \frac{\|z_j(\cdot)\|_{j \in \mathbb{N}_i}^2}{\|w_j(\cdot)\|_{j \in \mathbb{N}_i}^2} < \sup_{\{w_j(\cdot)\}_{j \in \mathbb{N}_{i-1}}} \frac{\|z_j(\cdot)\|_{j \in \mathbb{N}_{i-1}}^2}{\|w_j(\cdot)\|_{j \in \mathbb{N}_{i-1}}^2},$$

for all $i \in \mathbb{N}_N$ (with the right hand side for $i = 1$ being the prespecified constant $\tilde{\gamma}$). It is easy to see that this condition can be encoded in the proposed decentralized design (62) by replacing its local constraint $\hat{\gamma}_i < \tilde{\gamma}$ by an alternative local constraint of the form $\hat{\gamma}_i < \tilde{\gamma}_i \triangleq \min_{j \in \mathbb{N}_{i-1}} \hat{\gamma}_j$.

5.2. Vehicular platoon merging and splitting

The proposed decentralized scheme in Theorem 3 is “compositional” due to two main reasons: (1) each of its iterations (decentralized steps) is independent of the total number of vehicles in the platoon (i.e., N), and (2) at any iteration, the designed controllers do not affect the already designed controllers at previous iterations (except for the minor update (63)). This compositionality property allows one to conveniently and efficiently append new vehicles to an existing platoon, and thus, it is extremely helpful in handling scenarios where multiple platoons are merged in sequence.

Moreover, the proposed decentralized scheme in Theorem 3 can be executed in any arbitrary order. In other words, we can start the decentralized design procedures from any arbitrary vehicle (not necessarily from Σ_1) and continue until all the vehicles in the platoon are covered. While the optimality and the conservativeness of the resulting decentralized designs may vary depending on the followed order of the vehicles (see also Remark 12), a unique implication of this quality is that it can be used to conveniently and efficiently add an external vehicle into an existing platoon in the middle (not necessarily at the end) of that platoon. Consequently, this flexibility can be used to handle scenarios where multiple platoons are merged in parallel (i.e., not necessarily in sequence).

On the other hand, the compositionality property of the proposed decentralized designs can also be exploited to split an existing platoon into multiple platoons. For example, consider a situation where the decentralized design procedure is executed

sequentially from Σ_1 to Σ_N , in that order. Now, if the local design problem (62) fails to find a feasible solution at some vehicle $\Sigma_i, i \in \mathbb{N}_N \setminus \{1, N\}$, we can start a new platoon from that vehicle Σ_i and resume the design procedure there on-wards. This splitting strategy is particularly useful when enforcing additional stricter constraints like the string stability conditions mentioned in Remark 13.

5.3. String stability properties of the platoon

We conclude this section by providing a formal proof of L_2 weak string stability (defined in Definition 4) for the closed-loop error dynamics of the platoon (shown in Fig. 3). In particular, this string stability result given in the following theorem stems from the local passivity and global finite-gain L_2 stability properties enforced for the platoon using local and global controller designs, respectively (e.g., via Theorem 3).

Theorem 4. *If the closed-loop error dynamics of the platoon is finite-gain L_2 -stable with some L_2 gain γ such that $\gamma^2 < \bar{\gamma}$ as in Theorems 1 or 3, then it is L_2 weakly string stable.*

Proof. The differential dissipativity inequality corresponding to the established L_2 -stability (equivalently, Y-EID (50)) property takes the form:

$$\dot{V}(e(t), e^*) \leq \bar{\gamma}^2 |w(t)|^2 - |z(t)|^2, \quad \forall t \geq 0, \quad (66)$$

where $e(t)$, $w(t)$ and $z(t)$ are the error state, disturbance input and performance output of the closed-loop error dynamics of the platoon, respectively, and $e^* = \mathbf{0}$ (see also Definition 1 and Remark 1). Assuming the initial state as $e(0) = \mathbf{0}$, upon integrating (66) and using the fact that $V(\mathbf{0}, \mathbf{0}) = 0$, we can obtain

$$V(e(\tau), \mathbf{0}) \leq \int_0^\tau \bar{\gamma}^2 |w(t)|^2 - |z(t)|^2 dt, \quad \forall \tau \geq 0. \quad (67)$$

Since $0 \leq V(e(\tau), \mathbf{0})$, for $\tau = \infty$, (67) implies $0 \leq \bar{\gamma}^2 \|w\|^2 - \|z\|^2$. Now, using the fact that $z(t) \triangleq e(t)$, we get

$$\|e\| \leq \bar{\gamma} \|w\|, \quad (68)$$

which implies that L_2 weakly string stability condition (15) in Definition 4 (with $\alpha_1(s) \triangleq \bar{\gamma}s$ and any arbitrary class $\mathcal{K}$ function $\alpha_2(s)$ independent of N) for the considered closed-loop error dynamics of the platoon. This completes the proof. ■

Remark 14. This paper focuses on developing a compositional control and communication topology co-design scheme for vehicular platoons to efficiently manage merging and splitting scenarios. We approached this problem from a network control perspective, and enforced string stability guarantees under a constant spacing policy as the primary safety constraints. While our current approach does not include strict safe-distance constraints, such constraints can be incorporated seamlessly in a decentralized manner using control barrier functions, which are well known for their integrability with existing controllers to ensure that required strict safety constraints are met (Lindemann & Dimarogonas, 2019; Xiao & Belta, 2019).

6. Simulation results

In this section, we verify our proposed co-design framework detailed in the previous sections using numerical simulations. We study two cases (i.e., centralized and decentralized) of platooning co-design with L_2 weak string stability guarantees. For each case, we consider a vehicular platoon containing ten vehicles; the first vehicle is viewed as the leader, and the rest are the followers.

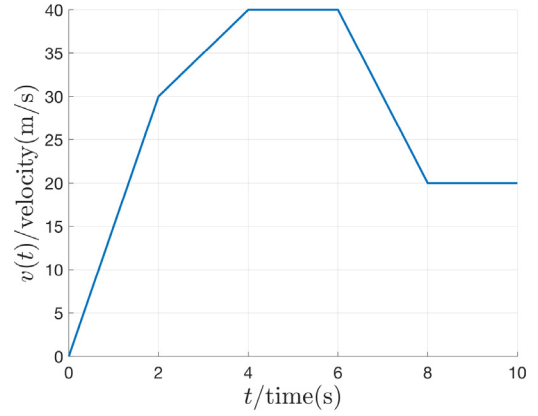


Fig. 5. Reference velocity of the leader.

Without loss of generality, we assume the platoon to be homogeneous and each vehicle in (16) is with the parameters $m_i = 1500$ kg, $\tau_i = 0.25$ s, $A_{f,i} = 2.2$ m², $\rho = 0.78$ kg/m³, $C_{d,i} = 0.35$ and $C_{r,i} = 0.067$, for any $i \in \mathbb{N}_9$. Besides, the desired separating distances between two successive vehicles and the length of a vehicle are selected as $\delta_{di} = 5$ m and $L_i = 2.5$ m, respectively. Furthermore, each external disturbance component of each vehicle is assumed to be independent and zero-mean Gaussian random distributed with a variance magnitude $\sim U(0, 0.1)$ ($U(0, 0.1)$ denotes the uniform random distribution in the interval $(0, 0.1)$). The initial locations of all vehicles in the platoon are assumed to be $x_i(0) = x_0 - i \times x_{im} - x_{iv}$, for all $i \in \mathbb{N}_9$, where $x_0 = 0$ is the position of the leader, $x_{im} = 5$ and $x_{iv} \sim U(-1, 1)$ are the mean and variance of the separation between two successive vehicles, respectively.

For the simulation of both considered cases, we assume the leader's reference velocity to be a piecewise linear function (as shown in Fig. 5):

$$v_1(t) : \begin{cases} 15t \text{ m/s}, & 0 \leq t \leq 2 \text{ s} \\ (5t + 20) \text{ m/s}, & 2 \leq t \leq 4 \text{ s} \\ 40 \text{ m/s}, & 4 \leq t \leq 6 \text{ s} \\ (-10t + 100) \text{ m/s}, & 6 \leq t \leq 8 \text{ s} \\ 20 \text{ m/s}. & 8 \leq t \leq 10 \text{ s} \end{cases} \quad (69)$$

For the co-design process, as a reference topology for the platoon (i.e., to select cost function coefficients in (51) and (62)), we use the reference topology shown in Fig. 6. Simulation results are generated by a simulator developed in MATLAB.³ For the considered two cases, the observed co-design solutions (topologies) are shown in Figs. 7(a) and 7(d), and the observed state profiles and tracking errors are shown in Figs. 8(a)–8(d) and Figs. 9(a)–9(d), respectively.

6.1. Case 1: Centralized platooning co-design

We first verify the centralized version of the proposed co-design method, i.e., the program (51) in Theorem 1 (with the program (61) in Theorem 2). Based on the above platoon parameters, our simulation results are outlined in Fig. 7(a) and Figs. 8(a)–8(d). Specifically, the platoon under the synthesized centralized communication topology is shown in Fig. 7(a), where the communication links are mostly between farther neighbors (not between immediate neighbors). This behavior results from the trade-off between the communication cost and the disturbance robustness. For example, the communication links from the

³ Publicly available in <https://github.com/NDZsong2/Longitudinal-Vehicular-Platoon-Simulator.git>

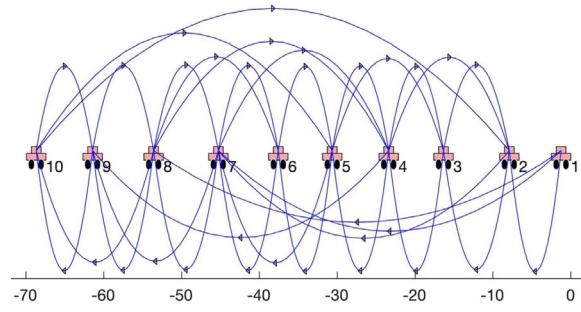


Fig. 6. Reference communication topology.

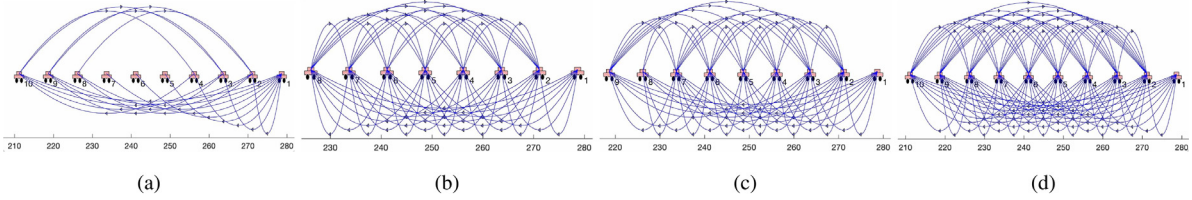


Fig. 7. (a) The platoon under the centralized controller (topology) with 10 vehicles; (b)–(d) An example of the merging process with the decentralized controller (topology), where the platoon size grows from 8 vehicles to 10 vehicles. Directed arrows represent the interconnection topology, where the upper ones are the communication edges from the follower to the predecessor and vice versa.

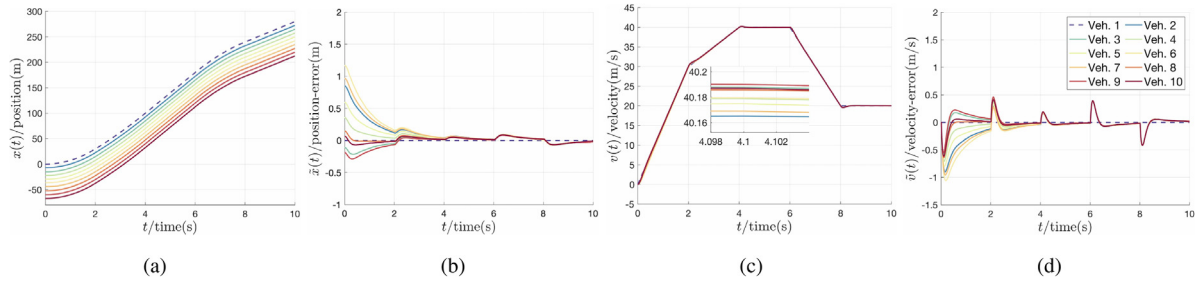


Fig. 8. (a) Position, (b) position tracking errors, (c) velocity, and (d) velocity tracking errors of the leading vehicle (dashed lines) and the following vehicles (solid lines) with our proposed centralized co-design method.

successive vehicles are beneficial for the disturbance robustness of the platoon. Still, the density of such links will significantly increase the communication cost. Therefore, several links between farther neighboring vehicles are retained, meaning these links are more cost-effective than those between immediate neighbors in terms of communication cost and control performance.

The observed optimal L_2 -gain from the centralized co-design is 1.7305. To quantify the tracking control result and string stability performance, the position and velocity tracking behaviors for the platoon under the centralized co-design are shown in Figs. 8(a)–8(d). These plots readily demonstrate that the position and velocity tracking of each vehicle can be well achieved. Specifically, each vehicle's real-time position is smoothly evolved, and no collisions (i.e., intersections between two position curves of two vehicles) occur. Besides, the position and velocity tracking errors are bounded, convergent, and non-increasing along the platoon. In particular, the centralized co-design solution enforces the L_2 weakly string stability (via Theorem 4), since the norms $\|\tilde{x}_i\|_\infty$ and $\|\tilde{v}_i\|_\infty$ are uniformly bounded by the maximum values 1.1668 and 1.0586 as seen from Figs. 8(b) and 8(d), respectively. In fact, it also satisfies the so-called Disturbance String Stability (DSS) (Besselink & Johansson, 2017), since the weak coupling condition $\max\{\sum_{j \in \mathbb{N}_9} |P_i^{-1} K_{ij}|\} = 4.6543 \times 10^{-4} < 1, \forall i \in \mathbb{N}_9$.

6.2. Case 2: Decentralized platooning co-design

For the vehicular platoon, we also verify the decentralized version of the proposed co-design method, i.e., the program in (62) in Theorem 3 (with the program (61) in Theorem 2). For the considered platoon, the synthesized decentralized communication topology is shown in Figs. 7(b)–7(d). In particular, these figures show scenarios where the 7th, 8th and 9th follower vehicles are being sequentially added/dropped in the simulation, i.e., where the program (62) in Theorem 3 is iteratively solved for $i = 7, 8$ and 9, respectively.

An interesting phenomenon that can be observed in Figs. 7(b)–7(d) is the appearance/disappearance of communication links when vehicles joining/leaving the platoon. This practically makes sense because vehicles joining/leaving will correspondingly change the disturbance robustness properties of the entire platoon. Another phenomenon that can be observed in Fig. 7(d) is the increased number of followers–followers communication links as compared to the centralized case in Fig. 7(a). This is likely caused by the decentralized nature of the co-design process (62), in particular, since the disturbance robustness capability is incrementally developed, it leads to a more conservative result than in the centralized case (i.e., it leads to a topology with more communication links compared to that of the centralized case,

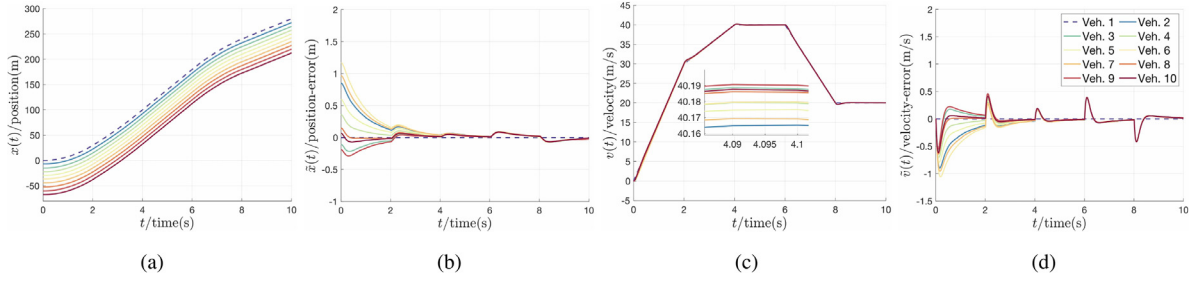


Fig. 9. (a) Position, (b) position tracking errors, (c) velocity, and (d) velocity tracking errors of the leading vehicle (dashed lines) and the following vehicles (solid lines) with our proposed decentralized co-design method.

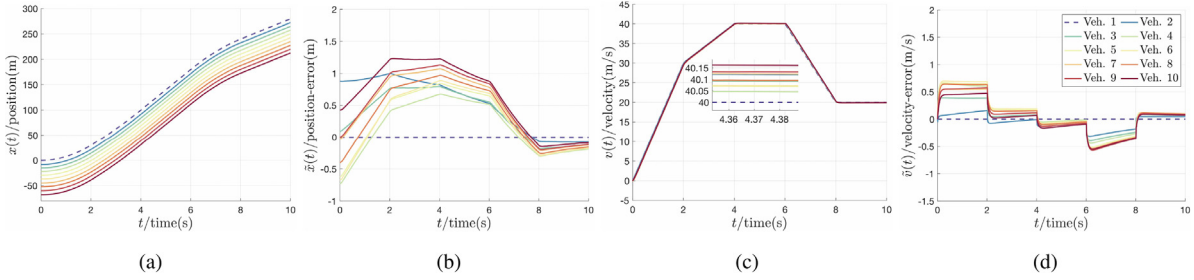


Fig. 10. (a) Position, (b) position tracking errors, (c) velocity, and (d) velocity tracking errors of the leading vehicle (dashed lines) and the following vehicles (solid lines) with the PID controller in Dasgupta et al. (2017).

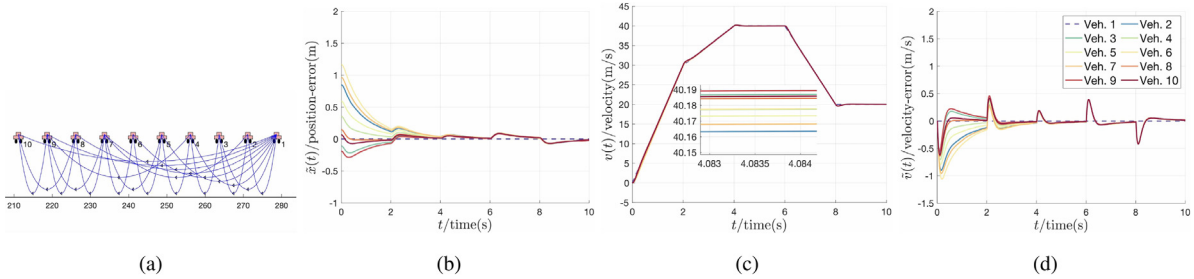


Fig. 11. (a) The platoon under the topology with only links from front to back (e.g., conventional CACC applications); (b) position tracking errors, (c) velocity, and (d) velocity tracking errors of the leading vehicle (dashed lines) and the following vehicles (solid lines) with decentralized method.

despite both cases providing normal platooning behaviors).

In other words, our decentralized solution can be restrictive in terms of communication resource utilization. Theoretically, our decentralized solution is conservative since the satisfaction of the LMI in (62) can also make its centralized version (51) hold, but not vice versa. Moreover, unlike in the centralized co-design (51) where the global configuration of the network is known, in decentralized co-design (62), the eventual global configuration of the network is not known.

The observed optimal L_2 -gain from the decentralized co-design is 1.9144. This increased L_2 -gain value (with respect to that in the centralized case: 1.7305) is indicative of the said conservativeness of the decentralized co-design result. To quantify the tracking control result and string stability performance, the position and velocity tracking behaviors for the platoon under the decentralized co-design are shown in Figs. 9(a)–9(d). The tracking behaviors are mostly similar to those in the centralized version. In particular, the decentralized co-design solution enforces the L_2 weak string stability (via Theorem 4) since the position and velocity tracking errors $\|\tilde{x}_i\|_\infty$ and $\|\tilde{v}_i\|_\infty$ are uniformly bounded by the maximum values 1.1668 and 1.0588,

as seen from Figs. 9(b) and 9(d), respectively. In fact, it also satisfies the DSS constraints since the weak coupling condition $\max\{\sum_{i \in \mathbb{N}_g} |P_i^{-1} K_{ij}|\} = 0.7361 < 1, \forall i \in \mathbb{N}_g$. Notice that these bounding values are larger than those observed in the centralized case. This is intuitive as more communication links will increase the available avenues to propagate the errors and disturbances over the platoon, thus weakening its string stability.

To show the compositionality of our proposed decentralized co-design method, during vehicular merging and splitting, we can use the topologies observed in the merging process illustrated in Fig. 7(b)–7(d). In particular, we observe that the inner portion of the topology (i.e., the topology involving only the follower vehicles 1 – 7) remains the same during the merging process of the 8th and 9th follower vehicles. This can also be shown using the corresponding controller gains to the inner portion of the topology.

Remark 15. Based on the above comparison between centralized and decentralized co-design, the centralized co-design outperforms the decentralized co-design in many aspects except for

being compositional. This is due to the centralized co-design having the knowledge of the entire network's configuration. Besides, the total execution time for the centralized co-design process (to obtain the result in Fig. 7(a)) is 1.35s, while it takes 3.24s for the entire decentralized co-design process (to obtain the result in Fig. 7(d)). The additional execution time used by the decentralized co-design process is due to the information exchanges required between subsystems. However, the compositionality property of the decentralized co-design process enables handling vehicular merging and splitting scenarios effectively without having to re-execute the co-design process for the current platoon. This is clear from the observed execution times for the decentralized co-design at subsequent follower vehicles: 0.20s, 0.20s, 0.24s, 0.34s, 0.43s, 0.38s, 0.53s, 0.55s and 0.37s (totaling to 3.24s), whereas the observed execution times for the centralized co-design when each subsequent vehicle is introduced: 0.20s, 0.44s, 0.49s, 0.55s, 0.73s, 0.82s, 0.93s, 1.09s and 1.35s.

6.3. Comparison study

To better illustrate the effectiveness of our method, we compare our results to an existing state-of-the-art PID control method (Dasgupta et al., 2017) for vehicular merging/splitting mentioned in Section 1. For this comparison, we consider the same platooning setup as mentioned above, but each following vehicle is controlled by a PID controller under the predecessor leader following type topology as is used in Dasgupta et al. (2017). We tune the PID controller gains using the `fmincon` function in MATLAB by optimizing the overall tracking error. The optimal controller parameters we obtained are: $k_p = 4544.25$, $k_i = 1133.22$, $k_d = 83.82$, and the observed simulation results are shown in Fig. 10.

From the comparison to the results in Fig. 9 observed using our proposed co-design method, it can be readily seen that while both methods can ensure the normal behavior of the platoon (position and velocity tracking as seen in Figs. 9(a), 9(c) and Figs. 10(a), 10(c) and string stability as seen in Figs. 9(b), 9(d) and Figs. 10(b), 10(d)), our proposed decentralized solution outperforms in terms of the position and velocity tracking errors, which is obviously observed from the comparison between Figs. 9(b), 9(d) and Figs. 10(b), 10(d). Particularly, the position tracking errors in Fig. 9(b) obtained by our proposed method converges rapidly to zero while those by the PID method in Dasgupta et al. (2017) can only guarantee the convergence around some neighborhood of zero with large offsets. For velocity tracking, results in Fig. 10(d) by the PID method in Dasgupta et al. (2017) undergo large deviation from zero, especially in the intervals of 0–2s and 6–8s. Another drawback of the existing PID controller is due to the fact that the tuned control parameters have to be retuned after vehicular merging/splitting. For example, when the last vehicle leaves the platoon, the resulting optimal PID control parameters become $k_p = 2537.77$, $k_i = 2149.41$, $k_d = 101.14$, which implies that the controller is not compositional.

Note that there are unusual links from the back to the front of the platoon appearing in both centralized and decentralized co-design solutions. This observation may contradict with typical topologies used for platooning control, e.g., in the applications of cruise control. To show the benefit of these links, we further compare our decentralized co-design results with a scenario where only the links from the front to the back of the platoon are used. In this case, the observed results are shown in Fig. 11. Even though the tracking behaviors shown in Figs. 11(b)–11(d) are similar and comparable to those in Figs. 9(b)–9(d), the resulting optimal L_2 -gain is 2.0199 for the case of Fig. 11 (with respect to that in the case of Fig. 9: 1.9144). To quantitatively illustrate the overall performance, we compare a total cost metric

of the communication and disturbance robustness, defined as $J \triangleq \sum_{i,j \in \mathbb{N}_N} |K_{ij}| + \gamma$. The observed J values for the case of Fig. 11(a) is $J = 2.2592$ while that for the case of Fig. 9 is $J = 2.1179$. This indicates that the topology in Fig. 7(d) obtained by the proposed decentralized co-design (with links from the back to the front of the platoon), is reasonable and cost-effective.

7. Conclusion

In this paper, we proposed a dissipativity-based co-design framework for vehicular platoons' distributed controller and communication topology. This is a well-motivated problem and of immense practical importance, especially when a platoon is splitting into multiple platoons or two or more platoons merging. Our proposed design method is distributed since the controller for each vehicle only uses the information from the neighboring vehicles. Furthermore, our solution is compositional in that the distributed controller can be composed incrementally, and the controllers (topology weights) of the remaining vehicles do not need to be re-designed when vehicles join/leave the platoon. We showed that the proposed co-design method ensures the L_2 stability of the closed-loop platooning system, which further implies the L_2 weak string stability. Simulation results illustrated the applicability of the proposed method. In particular, it was observed that both centralized and decentralized co-design frameworks provide feasible solutions, but the decentralized case requires more communication links than that in the centralized case. Some notable future research directions are: (1) enforcing a constant headway policy as opposed to a constant spacing policy, (2) enforcing strict safe-distance constraints, (3) addressing parametric uncertainties that may hinder feedback linearization via integrating dissipativity-based integral quadratic constraints, (4) handling vehicular platoons in multiple parallel lanes with string stability guarantees and (5) improving computational complexity of the design processes from $O(N^2)$ to $O(N \log(N))$.

Appendix

A. Proof of Proposition 1

Proof. Under a constant input $u(t) = u^*$, the state $x(t) = x^*$ is an equilibrium state of (2) if $Ax^* + Bu^* = 0$, and the corresponding equilibrium output is $y(t) = y^* \triangleq Cx^* + Du^*$.

(Necessity) Consider the candidate storage function $V(x, x^*) = (x - x^*)^T P(x - x^*)$ where $P > 0$ which leads to

$$\dot{V}(x, x^*) = (x - x^*)^T P(Ax + Bu) + (Ax + Bu)^T P(x - x^*).$$

Now, since $Ax + Bu = Ax^* + Bu^*$ where $\bar{x} \triangleq x - x^*$ and $\bar{u} \triangleq u - u^*$, we can simplify the above $\dot{V}(x, x^*)$ expression as

$$\dot{V}(x, x^*) = \bar{x}^T \mathcal{H}(PA)\bar{x} + \bar{x}^T P\bar{B}\bar{u} + \bar{u}^T B^T P\bar{x}. \quad (70)$$

On the other hand, according to Definition 2, for (2) to be X-EID,

$$\dot{V}(x, x^*) \leq \begin{bmatrix} u - u^* \\ y - y^* \end{bmatrix}^T \begin{bmatrix} X^{11} & X^{12} \\ X^{21} & X^{22} \end{bmatrix} \begin{bmatrix} u - u^* \\ y - y^* \end{bmatrix}.$$

However, since $y - y^* = C\bar{x} + D\bar{u}$, the above inequality can be simplified into the form:

$$\begin{aligned} \dot{V}(x, x^*) &\leq \bar{x}^T (C^T X^{22} C) \bar{x} + \bar{x}^T (C^T X^{21} + C^T X^{22} D) \bar{u} \\ &\quad + \bar{u}^T (X^{12} C + D^T X^{22} C) \bar{x} \\ &\quad + \bar{u}^T (X^{11} + \mathcal{H}(X^{12} D) + D^T X^{22} D) \bar{u}. \end{aligned} \quad (71)$$

Combining (70) and (71), we can obtain the condition:

$$\begin{bmatrix} \bar{x} \\ \bar{u} \end{bmatrix}^\top W \begin{bmatrix} \bar{x} \\ \bar{u} \end{bmatrix} \geq 0,$$

where W is the block matrix seen in the LMI in Proposition 1. This condition holds whenever $W \geq 0$, irrespective of x, u, x^* and u^* . Thus, (2) is X -EID if there exists $P > 0$ such that $W \geq 0$.

(Sufficiency) Assume that we can find some $P > 0$ such that $W \geq 0$ irrespective of x, u, x^* and u^* . This implies that the storage function $V(x, x^*) \triangleq (x - x^*)^\top P (x - x^*)$ satisfies the conditions stated in Definitions 1 and 2 (i.e., the conditions for (2) to be X -EID, see also de S. Madeira (2022) for a complete treatment). ■

B. Proof of Corollary 1

Proof. The proof starts by applying Proposition 1 to the considered system (4). Basically, in (3), we need to change the variables: $A \rightarrow A + BL$, $B \rightarrow \mathbf{I}$, $C \rightarrow \mathbf{I}$ and $D \rightarrow \mathbf{0}$, which gives

$$\begin{bmatrix} -\mathcal{H}(P(A + BL)) + X^{22} & -P + X^{21} \\ \star & X^{11} \end{bmatrix} \geq 0.$$

Now, using the congruence principle (in particular, by pre- and post-multiplying by $\text{diag}(P^{-1}, \mathbf{I})$), we can obtain

$$\begin{bmatrix} -\mathcal{H}((A + BL)P^{-1}) + P^{-1}X^{22}P^{-1} & -I + P^{-1}X^{21} \\ \star & X^{11} \end{bmatrix} \geq 0.$$

Changing variables via $K \triangleq LP^{-1}$ and $P \triangleq P^{-1}$ then gives

$$\begin{bmatrix} -\mathcal{H}(AP + BK) + PX^{22}P & -I + PX^{21} \\ \star & X^{11} \end{bmatrix} \geq 0.$$

Finally, using $X^{22} < 0$ and applying the Schur complements theory, we can obtain the LMI condition in (5). ■

C. Proof of Proposition 2

Proof. As each subsystem $\Sigma_i, i \in \mathbb{N}_N$ is X_i -EID, there exists a storage function $V_i(x_i, x_i^*)$ such that

$$\dot{V}_i(x_i, x_i^*) \leq \begin{bmatrix} u_i - u_i^* \\ y_i - y_i^* \end{bmatrix}^\top X_i \begin{bmatrix} u_i - u_i^* \\ y_i - y_i^* \end{bmatrix}, \quad (72)$$

for all $(x_i, x_i^*, u_i) \in \mathbb{R}^{n_i} \times \mathcal{X}_i \times \mathbb{R}^{q_i}$ and $i \in \mathbb{N}_N$. Now, for the networked system Σ , consider the candidate storage function $V(x, x^*) \triangleq \sum_{i \in \mathbb{N}_N} p_i V_i(x_i, x_i^*)$, which, with (72), leads to

$$\dot{V}(x, x^*) \leq \begin{bmatrix} u - u^* \\ y - y^* \end{bmatrix}^\top X_p \begin{bmatrix} u - u^* \\ y - y^* \end{bmatrix},$$

for all $(x, x^*, u) \in \mathbb{R}^n \times \mathcal{X} \times \mathbb{R}^q$. Consequently, for the networked system Σ to be $\mathbf{Y}$ -EID, we require

$$\begin{bmatrix} u - u^* \\ y - y^* \end{bmatrix}^\top X_p \begin{bmatrix} u - u^* \\ y - y^* \end{bmatrix} \leq \begin{bmatrix} w - w^* \\ z - z^* \end{bmatrix}^\top Y \begin{bmatrix} w - w^* \\ z - z^* \end{bmatrix}. \quad (73)$$

Besides, using the interconnection relationship (7) and equilibrium properties of the networked system Σ , we can write

$$\begin{bmatrix} u - u^* \\ y - y^* \end{bmatrix} = \begin{bmatrix} M_{uy} & M_{uw} \\ \mathbf{I} & \mathbf{0} \end{bmatrix} \begin{bmatrix} y - y^* \\ w - w^* \end{bmatrix}, \quad (74)$$

$$\begin{bmatrix} w - w^* \\ z - z^* \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ M_{zy} & M_{zw} \end{bmatrix} \begin{bmatrix} y - y^* \\ w - w^* \end{bmatrix}.$$

Finally, by applying (74) in (73) and re-arranging the terms, we can obtain the LMI in (8). This completes the proof. ■

D. Proof of Proposition 3

Proof. First, we restate the condition for the networked system Σ to be $\mathbf{Y}$ -EID (i.e., (8) from Proposition 2) as

$$\begin{bmatrix} M_{uy} & M_{uw} \\ M_{zy} & M_{zw} \end{bmatrix}^\top \begin{bmatrix} X_p^{11} & \mathbf{0} \\ \mathbf{0} & -Y^{22} \end{bmatrix} \begin{bmatrix} M_{uy} & M_{uw} \\ M_{zy} & M_{zw} \end{bmatrix} + \begin{bmatrix} M_{uy}^\top X_p^{12} + X_p^{21} M_{uy} + X_p^{22} & X_p^{21} M_{uw} - M_{zy}^\top Y^{21} \\ M_{uw}^\top X_p^{12} - Y^{12} M_{zy} & -(M_{zw}^\top Y^{21} + Y^{12} M_{zw} + Y^{11}) \end{bmatrix} \leq 0. \quad (75)$$

Note that, under Assumption 1, when $X_i^{11} > 0, \forall i \in \mathbb{N}_N$, we get $\begin{bmatrix} X_p^{11} & \mathbf{0} \\ \mathbf{0} & -Y^{22} \end{bmatrix} > 0$. Therefore, Welikala et al. (2023, Lm. 1) is applicable to obtain an equivalent condition to (75), which, under the change of variables $L_{uy} \triangleq X_p^{11} M_{uy}$ and $L_{uw} \triangleq X_p^{11} M_{uw}$, leads to (9). On the other hand, under Assumption 1, when $X_i^{11} < 0, \forall i \in \mathbb{N}_N$, Welikala et al. (2023, Lm. 2) is applicable to obtain an alternative equivalent condition to (75) as (10) (see also (Welikala, Lin, & Antsaklis, 2022c, Rm. 14)). ■

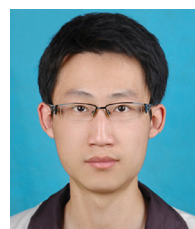
References

- Antsaklis, Panos J., & Michel, Anthony N. (2006). *Linear systems*. Birkhauser, <http://dx.doi.org/10.1007/0-8176-4435-0>.
- Arcak, Murat (2022). Compositional design and verification of large-scale systems using dissipativity theory. *IEEE Control Systems Magazine*, 42(2), 51–62. <http://dx.doi.org/10.1109/MCS.2021.3139721>.
- Besselink, Bart, & Johansson, Karl H. (2017). String stability and a delay-based spacing policy for vehicle platoons subject to disturbances. *IEEE Transactions on Automatic Control*, 62(9), 4376–4391.
- Boyd, Stephen, El Ghaoui, Laurent, Feron, Eric, & Balakrishnan, Venkataramanan (1994). *Linear matrix inequalities in system and control theory*. SIAM, <http://dx.doi.org/10.1137/1.9781611970777>.
- Chen, Na, Wang, Meng, Alkim, Tom, & Van Arem, Bart (2018). A robust longitudinal control strategy of platoons under model uncertainties and time delays. *Journal of Advanced Transportation*, 2018.
- Chou, Fang-Chieh, Tang, Shu-Xia, Lu, Xiao-Yun, & Bayen, Alexandre (2019). Backstepping-based time-gap regulation for platoons. In *Proc. of American control conference* (pp. 730–735).
- Dasgupta, Soumya, Raghuraman, Varunkumar, Choudhury, Apratim, Teja, T. Nagacharan, & Dauwels, Justin (2017). Merging and splitting maneuver of platoons by means of a novel PID controller. In *2017 IEEE symposium series on computational intelligence* (pp. 1–8). IEEE.
- de S. Madeira, Diego (2022). Necessary and sufficient dissipativity-based conditions for feedback stabilization. *IEEE Transactions on Automatic Control*, 67(4), 2100–2107. <http://dx.doi.org/10.1109/TAC.2021.3074850>.
- di Bernardo, Mario, Falcone, Paolo, Salvi, Alessandro, & Santini, Stefania (2015). Design, analysis, and experimental validation of a distributed protocol for platooning in the presence of time-varying heterogeneous delays. *IEEE Transactions on Control Systems Technology*, 24(2), 413–427.
- Feng, Shuo, Zhang, Yi, Li, Shengbo Eben, Cao, Zhong, Liu, Henry X, & Li, Li (2019). String stability for vehicular platoon control: Definitions and analysis methods. *Annual Reviews in Control*, 47, 81–97.
- Fiengo, Giovanni, Lui, Dario Giuseppe, Petrillo, Alberto, Santini, Stefania, & Tufo, Manuela (2019). Distributed robust PID control for leader tracking in uncertain connected ground vehicles with V2V communication delay. *IEEE/ASME Transactions on Mechatronics*, 24(3), 1153–1165.
- Goli, Mohammad, & Eskandarian, Azim (2019). MPC-based lateral controller with look-ahead design for autonomous multi-vehicle merging into platoon. In *Proc. of American control conference* (pp. 5284–5291).
- Guo, Xianggui, Wang, Jianliang, Liao, Fang, & Teo, Rodney Swee Huat (2016). Distributed adaptive integrated-sliding-mode controller synthesis for string stability of vehicle platoons. *IEEE Transactions on Intelligent Transportation Systems*, 17(9), 2419–2429.
- Guo, Ge, & Wen, Shixi (2015). Communication scheduling and control of a platoon of vehicles in VANETs. *IEEE Transactions on Intelligent Transportation Systems*, 17(6), 1551–1563.
- Hao, He, & Barooah, Prabir (2012). On achieving size-independent stability margin of vehicular lattice formations with distributed control. *IEEE Transactions on Automatic Control*, 57(10), 2688–2694.
- Hao, He, Barooah, Prabir, & Mehta, Prashant G. (2011). Stability margin scaling laws for distributed formation control as a function of network structure. *IEEE Transactions on Automatic Control*, 56(4), 923–929.

- Herman, Ivo, Martinec, Dan, Hurák, Zdeněk, & Šebek, Michael (2014). Nonzero bound on Fiedler eigenvalue causes exponential growth of H-infinity norm of vehicular platoon. *IEEE Transactions on Automatic Control*, 60(8), 2248–2253.
- Huang, Darong, Li, Shaoqian, Zhang, Zhenyuan, Liu, Yang, & Mi, Bo (2022). Design and analysis of longitudinal controller for the platoon with time-varying delay. *IEEE Transactions on Intelligent Transportation Systems*, 23(12), 23628–23639.
- Ji, Xuewu, He, Xiangkun, Lv, Chen, Liu, Yahui, & Wu, Jian (2018). Adaptive-neural-network-based robust lateral motion control for autonomous vehicle at driving limits. *Control Engineering Practice*, 76, 41–53.
- Jia, Dongyao, Lu, Kejie, Wang, Jianping, Zhang, Xiang, & Shen, Xuemin (2015). A survey on platoon-based vehicular cyber-physical systems. *IEEE Communications Surveys & Tutorials*, 18(1), 263–284.
- Kavathekar, Pooja, & Chen, YangQuan (2011). Vehicle platooning: A brief survey and categorization. In *International design engineering technical conferences and computers and information in engineering conference*: vol. 54808, (pp. 829–845).
- Li, B., & Yu, F. (2010). Design of a vehicle lateral stability control system via a fuzzy logic control approach. *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering*, 224(3), 313–326.
- Lindemann, Lars, & Dimarogonas, Dimos V. (2019). Decentralized control barrier functions for coupled multi-agent systems under signal temporal logic tasks. In *Proc. of 18th European control conf.* (pp. 89–94). <http://dx.doi.org/10.23919/ECC.2019.8796109>.
- Liu, Di, Besselink, Bart, Baldi, Simone, Yu, Wenwu, & Trentelman, Harry L. (2021). An adaptive disturbance decoupling perspective to longitudinal platooning. *IEEE Control Systems Letters*, 6, 668–673.
- Lofberg, J. (2004). YALMIP: A Toolbox for Modeling and Optimization in MATLAB. In *Proc. of IEEE intl. conf. on robotics and automation* (pp. 284–289). <http://dx.doi.org/10.1109/CACSD.2004.1393890>.
- Mirabilio, Marco, Iovine, Alessio, De Santis, Elena, Di Benedetto, Maria Domenica, & Pola, Giordano (2021). Scalable mesh stability of nonlinear interconnected systems. *IEEE Control Systems Letters*, 6, 968–973.
- Morales, América, & Nijmeijer, Henk (2016). Merging strategy for vehicles by applying cooperative tracking control. *IEEE Transactions on Intelligent Transportation Systems*, 17(12), 3423–3433.
- Morbidi, Fabio, Colaneri, Patrizio, & Stanger, Thomas (2013). Decentralized optimal control of a car platoon with guaranteed string stability. In *European control conference* (pp. 3494–3499).
- Orki, Omer, & Arogeti, Shai (2019). Control of mixed platoons consist of automated and manual vehicles. In *2019 IEEE international conference on connected vehicles and expo* (pp. 1–6). IEEE.
- Pirani, Mohammad, Baldi, Simone, & Johansson, Karl Henrik (2022). Impact of network topology on the resilience of vehicle platoons. *IEEE Transactions on Intelligent Transportation Systems*, 23(9), 15166–15177.
- Ploeg, Jeroen, Shukla, Dipan P., Van De Wouw, Nathan, & Nijmeijer, Henk (2013). Controller synthesis for string stability of vehicle platoons. *IEEE Transactions on Intelligent Transportation Systems*, 15(2), 854–865.
- Ploeg, Jeroen, Van De Wouw, Nathan, & Nijmeijer, Henk (2013). Lp string stability of cascaded systems: Application to vehicle platooning. *IEEE Transactions on Control Systems Technology*, 22(2), 786–793.
- Rahnama, Arash, Xia, Meng, & Antsaklis, Panos J. (2016). QSR-dissipativity and passivity analysis of event-triggered networked control cyber-physical systems. (p. 1607.00553). arXiv e-prints, URL: <https://arxiv.org/abs/1607.00553>.
- Song, Zihao, Welikala, Shirantha, Antsaklis, Panos J., & Lin, Hai (2022). Distributed adaptive backstepping control for vehicular platoons with mismatched disturbances using vector string Lyapunov functions. (p. 2207.02101). arXiv e-prints, [arXiv:2207.02101](https://arxiv.org/abs/2207.02101), URL: <http://arxiv.org/abs/2207.02101>.
- Song, Zihao, Welikala, Shirantha, Antsaklis, Panos J., & Lin, Hai (2023). Distributed adaptive backstepping control for vehicular platoons with mismatched disturbances using vector string Lyapunov functions. In *Proc. of American control conf.* (pp. 1–6).
- Sontag, Eduardo D., & Wang, Yuan (1995). On characterizations of the input-to-state stability property. *Systems & Control Letters*, 24(5), 351–359.
- Sun, Chuangchuang, Welikala, Shirantha, & Cassandras, Christos G. (2020). Optimal composition of heterogeneous multi-agent teams for coverage problems with performance bound guarantees. *Automatica*, 117, Article 108961. <http://dx.doi.org/10.1016/j.automatica.2020.108961>.
- Syed, Ali, Yin, George, Pandya, Abhilash, Zhang, Hongwei, et al. (2012). Coordinated vehicle platoon control: Weighted and constrained consensus and communication network topologies. In *Proc. of 51st conference on decision and control* (pp. 4057–4062).
- Tavan, Nematollah, Tavan, Mehdi, & Hosseini, Rana (2015). An optimal integrated longitudinal and lateral dynamic controller development for vehicle path tracking. *Latin American Journal of Solids and Structures*, 12, 1006–1023.
- Tibshirani, Robert (1996). Regression shrinkage and selection via the Lasso. *Journal of the Royal Statistical Society. Series B. Statistical Methodology*, 58(1), 267–288.
- Wang, Yan, Su, Rong, & Wang, Bohui (2022). Optimal control of interconnected systems with time-correlated noises: Application to vehicle platoon. *Automatica*, 137, Article 110018.
- Welikala, Shirantha, Lin, Hai, & Antsaklis, Panos (2022a). A generalized distributed analysis and control synthesis approach for networked systems with arbitrary interconnections. In *Proc. of 30th mediterranean conf. on control and automation* (pp. 803–808). <http://dx.doi.org/10.1109/MED54222.2022.9837162>.
- Welikala, Shirantha, Lin, Hai, & Antsaklis, Panos J. (2022b). A generalized distributed analysis and control synthesis approach for networked systems with arbitrary interconnections. (p. 2204.09756). arXiv e-prints, [arXiv:2204.09756](https://arxiv.org/abs/2204.09756), URL: <http://arxiv.org/abs/2204.09756>.
- Welikala, Shirantha, Lin, Hai, & Antsaklis, Panos J. (2022c). Centralized and decentralized techniques for analysis and synthesis of non-linear networked systems. (p. 2209.14552). ArXiv e-prints, [arXiv:2209.14552](https://arxiv.org/abs/2209.14552) URL: <http://arxiv.org/abs/2209.14552>.
- Welikala, Shirantha, Lin, Hai, & Antsaklis, Panos (2022d). On-line estimation of stability and passivity metrics. In *Proc. of 61st IEEE conf. on decision and control* (pp. 267–272).
- Welikala, Shirantha, Lin, Hai, & Antsaklis, Panos J. (2023). Non-linear networked systems analysis and synthesis using dissipativity theory. In *Proc. of American control conf.* (pp. 2951–2956). <http://dx.doi.org/10.23919/ACC55779.2023.10155851>.
- Wijnbergen, Paul, & Besselink, Bart (2020). Existence of decentralized controllers for vehicle platoons: On the role of spacing policies and available measurements. *Systems & Control Letters*, 145, Article 104796.
- Willems, Jan C. (1972). Dissipative dynamical systems part I: General theory. *Archive for Rational Mechanics and Analysis*, 45(5), 321–351. <http://dx.doi.org/10.1007/BF00276493>.
- Xiao, Wei, & Belta, Calin (2019). Control barrier functions for systems with high relative degree. In *Proc. of 58th conf. on decision and control* (pp. 474–479). <http://dx.doi.org/10.1109/CDC40024.2019.9029455>.
- Xiao, Lingyun, & Gao, Feng (2011). Practical string stability of platoon of adaptive cruise control vehicles. *IEEE Transactions on Intelligent Transportation Systems*, 12(4), 1184–1194.
- Yan, Yan, Du, Haiping, He, Defeng, & Li, Weihua (2022). Pareto optimal information flow topology for control of connected autonomous vehicles. *IEEE Transactions on Intelligent Vehicles*, 8(1), 330–343.
- Zheng, Yang, Li, Shengbo Eben, Wang, Jianqiang, Cao, Dongpu, & Li, Keqiang (2015). Stability and scalability of homogeneous vehicular platoon: Study on the influence of information flow topologies. *IEEE Transactions on Intelligent Transportation Systems*, 17(1), 14–26.
- Zhu, Yang, & Zhu, Feng (2018). Distributed adaptive longitudinal control for uncertain third-order vehicle platoon in a networked environment. *IEEE Transactions on Vehicular Technology*, 67(10), 9183–9197.



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